

Degree Project in ICT Innovation, EIT Digital - Data Science

Second cycle, 30 credits

Beyond Traditional Marketing Mix Modeling

An empirical evaluation of transformer-based approaches on aggregated marketing data

STEFANO ROMANO

Beyond Traditional Marketing Mix Modeling

An empirical evaluation of transformer-based approaches on aggregated marketing data

STEFANO ROMANO

Master's Programme, ICT Innovation, 120 credits
Date: June 23, 2026

Supervisor: Martin Trapp
Examiner: Henrik Böstrom
School of Electrical Engineering and Computer Science
Host company: Dema.ai
Swedish title: Bortom traditionell Marketing Mix-modellering
Swedish subtitle: En empirisk utvärdering av transformerbaserade metoder på aggregerad marknadsföringsdata

© 2026 Stefano Romano

Abstract

Marketing Mix Modeling (MMM) quantifies the impact of marketing activities on business outcomes using aggregated time-series data. Industry-standard Bayesian approaches impose parametric assumptions on temporal carryover effects and treat channels as independent, limiting their ability to capture heterogeneous response dynamics and inter-channel spillovers. Recent attention-based neural architectures offer greater flexibility, but have only been validated in simplified settings with few channels and rich embedding signals. Whether they can achieve predictive performance comparable to Bayesian MMMs while enabling richer structural analyses under realistic constraints, such as limited temporal observations, high channel dimensionality, and noisy data, remains an open question.

This thesis extends the Neural Network for Marketing Mix Modeling (NNN) framework with four architectural components for data-constrained environments: a hierarchical channel embedding that shares parameters across structurally related channels, independent per-channel temporal multi-layer perceptrons (MLPs) that replace the original shared network, a multi-scale attention mechanism that captures marketing effects at different time horizons simultaneously, and a Weibull-parametrized cross-channel interaction kernel that models lagged spillover effects with minimal parameters. The design follows an asymmetric complexity principle, allocating representational capacity where the learning signal is strong and constraining it where the signal is weak.

The architecture is evaluated on real-world marketing data from three e-commerce merchants and compared against a production-grade Bayesian MMM. The Bayesian model remains more accurate as a predictive tool overall. However, the neural model achieves comparable predictive performance in some settings, produces temporal response profiles that are substantially more heterogeneous than parametric alternatives, and captures cross-channel dynamics consistent with marketing funnel theory. A lag-zero overfitting tendency highlights the trade-off between flexibility and inductive bias in low-sample regimes.

These findings suggest that the two approaches are complementary: the Bayesian framework excels in robust prediction, while the neural model provides structural representations that enrich the understanding of marketing systems.

Keywords

Marketing Mix Modeling, Neural networks, Attention mechanism, Marketing attribution, Time-series analysis, Bayesian inference

Sammanfattning

Marketing Mix Modellering (MMM) är en metod för att mäta hur marknadsföringsinsatser påverkar affärsresultat med hjälp av aggregerade tidsseriedata. Befintliga branschstandardmodeller bygger på Bayesiansk statistik med fasta antaganden om hur marknadsföringseffekter avtar över tid och behandlar varje kanal oberoende av de andra.

Denna avhandling undersöker om neurala nätverk baserade på uppmärksamhetsmekanismer kan uppnå prediktiv prestanda som är jämförbar med Bayesianska MMM-modeller, samtidigt som de möjliggör rikare strukturella analyser under realistiska industriella villkor: få observationer (140–240 veckor), hög kanaldimensionalitet (9–14 kanaler) och brusiga aggregerade data. Arbetet bygger vidare på arkitekturen Neural Network for Marketing Mix Modeling (NNN) och introducerar fyra tillägg: en hierarkisk kanalrepresentation som delar parametrar mellan strukturellt relaterade kanaler, oberoende temporala nätverk per kanal som ersätter det ursprungliga delade nätverket, en flerskalig uppmärksamhetsmekanism som fångar marknadsföringseffekter över olika tidshorisonter, och en Weibull-parametriserad modell för fördröjda interaktioner mellan kanaler.

Systemet utvärderas på verkliga marknadsföringsdata från tre företag inom e-handel och jämförs med en produktionsmogen Bayesiansk MMM. Resultaten visar att den Bayesianska modellen är mer prediktivt noggrann överlag. Den neurala modellen uppnår dock jämförbar prediktiv prestanda i vissa fall, ger väsentligt rikare temporala profiler per kanal och fångar interaktioner mellan kanaler som överensstämmer med etablerad marknadsföringsteori.

Slutsatsen är att de två metoderna kompletterar varandra: den Bayesianska modellen lämpar sig bäst för robust prediktion, medan den neurala modellen bidrar med strukturella insikter som fördjupar förståelsen av marknadsföringssystem.

Nyckelord

Marketing Mix-modellering, Neurala nätverk, Uppmärksamhetsmekanism, Marknadsföringsattribuering, Tidsserieanalys, Bayesiansk inferens

Acknowledgments

Ringrazio Amin, Yun e Dima che mi hanno accompagnato in questo mio primo approccio al lavoro.

Ringrazio il professor Boström e il mio supervisor, Martin, per avermi seguito e aiutato durante il percorso.

Ringrazio i miei genitori che rimangono causa e fine delle mie decisioni. Grazie alle loro indicibili piccole e grandi azioni, che quotidianamente mi infondono fiducia e sicurezza nei miei mezzi, che mi permettono di avventurarmi lontano e sentirmi ad un passo casa.

Ringrazio gli Emanuele, Cecio, Marta e Marghe, che hanno condiviso con me tutte le battaglie della vita di questi anni, superandole una alla volta, tra chiacchiere, dubbi, sudore e sogni.

Ringrazio i ragazzi che mi fanno respirare quando torno a casa.

I would like to thank Amin, Yun, and Dima, who accompanied me in this first approach to work of mine.

Thanks to Professor Böstrom and my supervisor, Martin, for following me throughout this journey.

Thanks to my parents, who remain both the reason and the purpose behind my decisions, thanks to their inexpressable small and great acts that daily instill confidence in my own means, allowing me to adventure far and feel one step away from home.

Thanks to the Emanueles, Cecio, Marta and Marghe who have shared with me all the battles of life over these past years, overcoming them one by one, through conversations, doubts, sweat, and dreams.

Thanks to the guys who make me breathe when I get back home.

Stockholm, June 2026
Stefano Romano

Contents

Notation **xv**

1 Introduction **1**
1.1 Background . . . 1
1.2 Problem . . . 3
1.2.1 Problem definition in the industrial context . . . 4
1.2.2 Scientific and engineering issues . . . 5
1.3 Research Question . . . 5
1.4 Goals . . . 6
1.5 Research Methodology . . . 7
1.6 Delimitations . . . 8
1.7 Structure of the thesis . . . 8

2 Background **9**
2.1 Marketing Mix Model . . . 9
2.1.1 Bayesian Marketing Mix Models . . . 10
2.1.1.1 Model Specification . . . 10
2.1.1.2 Inference . . . 11
2.1.1.3 Limitations . . . 11
2.2 Neural and Attention-Based Approaches to Marketing Mix Modeling . . . 12
2.2.1 The NNN Framework . . . 12
2.2.1.1 Input Representation . . . 12
2.2.1.2 Transformer Architecture . . . 13
2.2.1.3 Factored Self-Attention . . . 13
2.2.1.4 Prediction Heads and Training . . . 14
2.2.1.5 Counterfactual Attribution . . . 14
2.2.1.6 Scope and Limitations . . . 15
2.3 Related Work . . . 15

2.3.1 Classical and Bayesian Extensions of Marketing Mix Modeling . . . 15
2.3.2 Causal and Structural Approaches to MMM . . . 16
2.3.3 Attention Mechanisms and Transformers for Time Series 17
2.3.4 Embedding Methods for Structured Data . . . 17
2.3.5 Multitask Learning . . . 17
2.3.6 Neural Approaches to Marketing Measurement . . . 18
2.3.7 Simulation and Evaluation Frameworks . . . 18

3 Methodology 19
3.1 Choice of research method . . . 19
3.1.1 Research paradigm . . . 19
3.1.2 Alternative methodologies considered . . . 20
3.1.3 Rationale for the chosen method . . . 21
3.2 Application of research method . . . 21
3.2.1 Research process . . . 21
3.2.2 Data collection . . . 21
3.2.3 Experimental design and planned measurements . . . 23
3.2.3.1 Test environment and model . . . 23
3.2.3.2 Training protocol and baseline comparison . 24
3.2.3.3 Hyperparameter selection . . . 24
3.2.4 Hardware and software environment . . . 25
3.3 Assessing reliability and validity of the data collected . . . 25
3.3.1 Validity of method . . . 26
3.3.2 Reliability of method . . . 26
3.3.3 Data validity . . . 26
3.3.4 Reliability of data . . . 26
3.4 Evaluation framework . . . 27
3.4.1 Predictive performance . . . 27
3.4.2 Model behavior and structural analysis . . . 28

4 Implementation 29
4.1 Model Architecture . . . 30
4.1.1 Data Representation . . . 30
4.1.1.1 Input Variables . . . 33
4.1.1.2 Temporal Aggregation . . . 33
4.1.2 Hierarchical Channel Embedding . . . 34
4.1.3 Multi-Scale Temporal Attention . . . 36
4.1.3.1 Multi-scale architecture . . . 37

4.1.3.2 Per-channel temporal attention 38
4.1.4 Cross-Channel Interaction Modeling 40
4.1.4.1 Static channel interaction matrix 40
4.1.4.2 Lag-dependent interaction effects 40
4.1.4.3 Weibull impulse response kernel 41
4.1.4.4 Channel interaction weights 42
4.1.4.5 Inductive bias and prior initialization 42
4.1.5 Factored attention combination 43
4.1.6 Prediction Heads and Baseline Decomposition 45
4.1.6.1 Per-channel media contribution 45
4.1.6.2 Baseline component 46
4.1.6.3 Final prediction 46
4.2 Training Objective 47
4.2.1 Primary prediction loss 47
4.2.2 Auxiliary multitask loss 47
4.2.3 Joint objective 48
4.2.4 Parameter regularization 48
4.2.5 Temporal diversity loss 48
4.2.6 Final loss function 49
4.3 Training Procedure 49
4.3.1 Training rationale 50
4.4 Hyperparameter Selection 50
4.4.1 Phase 1: Stabilization 51
4.4.2 Phase 2: Capacity 51
4.4.3 Phase 3: Baseline 51
4.5 Attribution Method 53
4.6 Implementation Summary 54

5 Results and Analysis **57**
5.1 Experimental Setup 57
5.2 Predictive Performance 58
5.2.1 Overall Accuracy 58
5.2.2 Prediction Over Time 59
5.3 Temporal Dynamics 60
5.3.1 Learned Temporal Profiles 60
5.3.2 Bayesian MMM Adstock 63
5.3.3 Comparison of Temporal Profiles 63
5.4 Cross-Channel Interactions 65
5.5 Attribution and Decomposition 67

5.5.1 Channel Contribution Breakdown 67
5.5.2 Channel Attribution Comparison 69
5.6 Reliability Analysis 70
5.7 Validity Analysis 71

6 Discussion **73**
6.1 Answering the Research Question 73
6.2 Flexibility vs Domain Knowledge 74
6.3 Practical Implications 75
6.4 Relation to Prior Work 76

7 Conclusions and Future work **79**
7.1 Conclusions 79
7.2 Limitations 80
7.3 Future work 81
7.4 Use of Generative AI 82
7.5 Ethical and Sustainability Considerations 82

References **83**

A Hyperparameter Optimization Details **87**
A.1 Phase 1: Stabilization 87
A.2 Phase 2: Capacity 87
A.3 Phase 3: Baseline 89

B Per-Merchant Temporal Analysis **91**
B.1 Merchant 1 91
B.2 Merchant 2 93
B.3 Merchant 3 95

List of Figures

4.1 Neural MMM architecture overview . . . 31
4.2 Data pipeline overview . . . 32
4.3 Hierarchical channel embedding . . . 35
4.4 Learned scale-weight distribution . . . 39
4.5 Cumulative adstock per channel . . . 39
4.6 Cross-channel influence over time . . . 43
4.7 Individual cross-channel interaction examples . . . 44
4.8 Chronological train/validation/test split . . . 50
4.9 Hyperparameter optimization: Phase 1 . . . 52
4.10 Hyperparameter optimization: Phase 2 . . . 52
4.11 Hyperparameter optimization: Phase 3 . . . 53

5.1 Neural model diagnostics, Merchant 3 . . . 59
5.2 Bayesian MMM diagnostics, Merchant 3 . . . 60
5.3 Neural model diagnostics, Merchant 1 . . . 61
5.4 Bayesian MMM diagnostics, Merchant 1 . . . 61
5.5 Neural model cumulative adstock . . . 62
5.6 Bayesian MMM cumulative adstock . . . 63
5.7 NNN vs MMM adstock comparison . . . 64
5.8 Cross-channel influence overview . . . 66
5.9 Within-platform funnel propagation . . . 66
5.10 Cross-platform delayed interaction . . . 67
5.11 Neural model temporal decomposition . . . 68
5.12 Bayesian MMM temporal decomposition . . . 68
5.13 Attribution: Channel2 PMax . . . 69
5.14 Attribution: Channel1 Sale . . . 70

A.1 Phase 1 optimization progress . . . 88
A.2 Phase 1 parameter distributions . . . 88
A.3 Phase 2 optimization progress . . . 89

A.4 Phase 2 parameter distributions 89
A.5 Phase 3 optimization progress 90
A.6 Phase 3 parameter distributions 90

B.1 Merchant 1 — Neural model adstock 92
B.2 Merchant 1 — Bayesian MMM adstock 92
B.3 Merchant 1 — NNN vs MMM comparison 93
B.4 Merchant 2 — Neural model adstock 94
B.5 Merchant 2 — Bayesian MMM adstock 94
B.6 Merchant 2 — NNN vs MMM comparison 95
B.7 Merchant 3 — Neural model adstock 95
B.8 Merchant 3 — Bayesian MMM adstock 96
B.9 Merchant 3 — NNN vs MMM comparison 96

List of Tables

4.1 Loss function coefficients 49
4.2 Fixed architectural parameters 51
4.3 Implementation summary 54

5.1 Dataset characteristics for the three evaluation merchants. . . . 58
5.2 Test-set predictive accuracy 58
5.3 Seed variance study for Merchant 1 (3 independent runs, same hyperparameters). 71

Notation

Symbol	Description
General indices and dimensions	
t, t'	Time indices (week)
c, v, c'	Channel indices
g	Geographical unit or merchant index
d	Day index (within a week) or embedding coordinate
j	Control variable or scale index
G	Number of geographical units or merchants
T	Number of time periods (weeks)
C	Number of marketing channels
D	Embedding dimension
L	Maximum adstock lag
Bayesian MMM (Chapter 2)	
y_t	Observed target at week t
μ_t	Expected response (mean of the likelihood)
σ	Observation noise standard deviation
b_t	Baseline (organic demand) term
$m_{t,c}$	Channel contribution of channel c at week t
$x_{t,c}$	Raw media spend of channel c at week t
$\tilde{x}_{t,c}$	Adstocked (carryover-transformed) spend
w_ℓ	Adstock kernel weight at lag ℓ
k_c, λ_c	Per-channel Weibull shape and scale parameters
β_c	Saturation asymptote for channel c
ρ_c	Half-saturation point for channel c
γ_j	Coefficient on control variable j
$z_{t,j}$	Control variable j at week t
s_t	Seasonal component (Fourier basis)

Symbol	Description
NNN framework (Chapter 2)	
$X \in \mathbb{R}^{G \times T \times C \times D}$	Input tensor (geo, time, channel, embedding)
$X^{(n)}$	Tensor after the n-th transformer layer ($X^{(0)} = X$)
N	Number of transformer layers
Θ	Full set of model parameters
a_{time}	Temporal attention score MLP
$\Delta_{t,t'}$	Normalized time lag, $(t - t')/\omega$
ω	Lookback window size (in weeks)
E_c	One-hot channel identifier, $\in \mathbb{R}^C$
$s_{t,t',c}$	Raw temporal attention score
τ	Softmax temperature
W_{time}	Temporal attention weight matrix
$\psi \in \mathbb{R}^{C \times C}$	Learnable channel interaction matrix
W_{chan}	Channel attention weight matrix
$O_{g,t,c,d}$	Factored attention output
$V_{g,t,c}$	Channel volume ($= \|X^{(N)}_{g,t,c}\|$)
$\bar{X}^{(N)}_{g,t,c}$	Unit-normalized embedding direction
$P(\cdot)$	Intent MLP (sigmoid output, learned saturation)
θ_c	Per-channel parameters of P
$\mathcal{L}_{\text{sales}}, \mathcal{L}_{\text{search}}$	Sales and search prediction losses
α	Task balancing coefficient
λ	L_1 regularization strength
Attribution_c	Counterfactual attribution of channel c
$\tilde{X}_c$	Input with channel c zeroed (counterfactual)
This thesis: data and normalization (Chapter 4)	
$x_{t,c} \in \mathbb{R}_{\geq 0}$	Weekly aggregated spend of channel c at week t
y_t	Primary target: Net Gross Profit at week t
$y_t^{(int)}$	Auxiliary target: impressions at week t
$z_t \in \mathbb{R}^{19}$	Control feature vector (lagged targets + calendar)
$\hat{y}_t, \hat{y}_t^{(int)}$	Predicted primary and auxiliary targets
s	Global normalization scale factor
$\tilde{X}, \tilde{y}$	Normalized inputs and target
This thesis: hierarchical channel embedding	
$E_{channel}(c)$	Learnable channel-identity embedding, $\in \mathbb{R}^D$
$E_{funnel}(c)$	Learnable funnel-stage embedding, $\in \mathbb{R}^D$

Symbol	Description
$E_{group}(c)$	Learnable channel-group embedding, $\in \mathbb{R}^D$
$f(\cdot)$	Spend projection function
W_{spend}	Weight vector of the spend projection, $\in \mathbb{R}^D$
b_{spend}	Bias term of the spend projection
$e(c,t)$	Final channel embedding for channel c at time t
This thesis: multi-scale temporal attention	
N	Number of temporal scales
ω_i	Temporal window of scale i (in weeks)
$s_c(t,t')$	Raw temporal attention score for channel c
MLP_c	Per-channel temporal MLP ($1 \to 16 \to 1$)
$W_{time}(t,c,t')$	Normalized temporal attention weight
τ_{temp}	Softmax temperature for temporal attention
H_i	Output representation of scale i
H	Combined multi-scale representation
α_i	Learned scale weight (softmax-normalized)
θ_i	Raw (pre-softmax) scale logit
This thesis: cross-channel interaction	
$S \in \mathbb{R}^{C \times C}$	Learnable static channel interaction matrix
$S_{c,v}$	Base interaction strength from channel v to c
$W_{channel}(c,v)$	Normalized channel attention weight
$\beta_{c,v}$	Amplitude of lagged cross-channel interaction
$K(\tau;k,\lambda)$	Weibull impulse response kernel
k	Shape parameter of the Weibull kernel
λ	Scale parameter of the Weibull kernel
$W(t,c,v,\tau)$	Combined factored attention weight
This thesis: prediction heads	
$effect_{t,c}$	Predicted media contribution of channel c at week t
w_c, b_c	Per-channel sales head parameters
$baseline_t$	Predicted organic baseline at week t
α_{base}	Learnable baseline scaling factor
Z_t	Baseline input features (lagged targets + calendar)
f_{base}	Baseline MLP
This thesis: loss function	
$\mathcal{L}_{sales}$	Primary MSE loss on Net Gross Profit
$\mathcal{L}_{int}$	Auxiliary MSE loss on impressions

Symbol	Description
$\mathcal{L}_{reg}$	L_1 parameter regularization
$\mathcal{L}_{div}$	Temporal diversity loss
$\mathcal{L}$	Total training loss
$\alpha_{sales}, \alpha_{int}$	Task weights in joint objective
λ	L_1 regularization strength
λ_{div}	Diversity loss weight
$\tilde{p}_c$	Scale-weighted temporal profile of channel c

Chapter 1

Introduction

This chapter introduces the research area, background, and motivation for the thesis. It defines the problem addressed from both an industrial and a scientific perspective, formulates the research question, and outlines the goals, methodology, delimitations, and structure of the work.

1.1 Background

Marketing effectiveness measurement has been a central concern in marketing science and applied econometrics for decades. The fundamental question of how marketing activities influence business outcomes has motivated extensive research into quantitative models capable of estimating the contribution of individual marketing channels to observed sales, revenue, or other performance indicators (Hanssens et al., 2001). This line of research, broadly referred to as market response modeling, provides the econometric foundations on which modern Marketing Mix Modeling is built (Hanssens et al., 2001).

Beyond its practical relevance, the MMM problem presents characteristics that make it a challenging and informative testbed for data science and machine learning research (Mulc et al., 2025; Gong et al., 2024). It requires modeling multivariate temporal dependencies under limited sample sizes, capturing non-linear and delayed interactions across channels without strong supervision signals, and producing interpretable outputs under high uncertainty. These constraints are shared by a broad class of applied time-series problems, making insights gained in the MMM domain relevant to the wider machine learning community (Lim et al., 2021; Spiliotis et al., 2022).

Across industries such as retail, consumer packaged goods, and financial

services, marketing investment decisions must be taken under limited observability, where user-level tracking and controlled experimentation are either unavailable or increasingly restricted by privacy regulations (Miller et al., 2026). As a result, organizations rely on aggregated data and statistical modeling to inform strategic decisions related to budget allocation and channel optimization. Within this context, Marketing Mix Modeling (MMM) has emerged as a widely adopted framework for estimating marketing contributions using aggregated time-series data (Chan and Perry, 2017; Jin et al., 2017). By operating at an aggregate level, MMM provides a privacy-preserving and operationally scalable alternative for marketing measurement.

In practice, MMM models typically include multiple marketing channels, each represented by historical spend or exposure variables, together with control variables capturing seasonality, trends, and external factors (Jin et al., 2017). A key challenge lies in capturing the temporal structure of marketing effects, including delayed responses, carryover effects, and saturation phenomena. Classical MMM approaches address these challenges through parametric transformations such as adstock functions (Broadbent, 1979) and response curves, often embedded within Bayesian regression frameworks (Chan and Perry, 2017; Ng et al., 2021).

Despite their practical success, several limitations of traditional MMMs have been identified in the literature. These include sensitivity to modeling assumptions, difficulties in representing complex cross-channel interactions (Naik and Raman, 2003), and limited flexibility in adapting to time-varying dynamics (Chan and Perry, 2017; Ng et al., 2021). Moreover, aggregated marketing data are typically characterized by a small number of time points relative to the dimensionality of the problem, which places strong constraints on model complexity and data efficiency (Guzman, 2024).

Recent advances in machine learning and sequence modeling have opened new possibilities for modeling temporal data with fewer explicit assumptions. In particular, attention-based neural architectures (Vaswani et al., 2017) have demonstrated strong capacity to capture long-range dependencies and non-linear relationships in sequential data, with successful applications in time-series forecasting (Lim et al., 2021). These developments raise the question of whether such models can be effectively applied to aggregated marketing data, and how they compare to established MMM methodologies under realistic constraints.

This thesis contributes to two intersecting research areas within data science and machine learning. First, it contributes to the study of *deep learning for time-series forecasting* (Lim et al., 2021), specifically

investigating how attention-based architectures perform in low-sample, high-noise multivariate settings where classical econometric methods have traditionally been preferred. Second, it contributes to the empirical understanding of *representation learning under data constraints*: whether learned temporal and cross-channel representations can provide modeling advantages over domain-informed parametric transformations when granular data are unavailable, a question of growing relevance as deep learning methods are applied to domains with strong prior structural knowledge (Mulc et al., 2025).

To address some of the limitations of traditional MMMs, recent research has explored more expressive modeling strategies, including simulation-based evaluation frameworks (Vaver and Zhang, 2017), time-varying parameter models (Ng et al., 2021), and causal formulations (Gong et al., 2024). In parallel, neural network–based approaches have been proposed to capture non-linear and mediated effects through learned representations and attention mechanisms (Mulc et al., 2025). These approaches suggest a shift from purely parametric modeling toward representation learning, while raising new questions regarding robustness, interpretability, and suitability for data-constrained settings.

This thesis builds on these developments by comparatively evaluating attention-based neural models and Bayesian MMMs under realistic industrial conditions. Detailed technical background and related work are presented in Chapter 2.

1.2 Problem

The problem addressed in this thesis arises at the intersection of industrial marketing practice and engineering research on data-driven decision-making. Organizations increasingly rely on quantitative models to guide marketing investments, yet the available data are often aggregated, noisy, and limited in temporal scope. Under these constraints, marketing effectiveness must be inferred indirectly, without access to user-level behavior or controlled experimental feedback.

Several findings in the existing literature motivate the need for investigating alternative modeling approaches. Chan and Perry (Chan and Perry, 2017) document key challenges faced by Bayesian MMMs in practice, including sensitivity to the choice of parametric transformations and difficulty in handling multicollinearity across channels. Ng et al. (Ng et al., 2021) show that marketing effects are often non-stationary, proposing time-varying

coefficient models to address this limitation, but note that such extensions increase complexity while remaining dependent on handcrafted functional forms. Gong et al. (Gong et al., 2024) demonstrate that ignoring the causal structure among marketing variables can lead to biased attribution estimates, yet their approach relies on strong assumptions about the data-generating process. More recently, Mulc et al. (Mulc et al., 2025) propose the NNN framework, showing that attention-based neural architectures can capture temporal and cross-channel effects without predefined parametric assumptions; however, their empirical validation is limited to settings with only two paid media channels and rich embedding signals, leaving open the question of whether such architectures generalize to more complex, data-constrained industrial environments.

Taken together, these findings point to a gap in the literature: while the limitations of parametric Bayesian MMMs are well documented, and neural alternatives have been proposed, there is limited empirical evidence on how these approaches compare under realistic industrial conditions with many channels, limited sample sizes, and sparse auxiliary signals.

At the same time, the broader machine learning literature on time-series forecasting has shown that the relative performance of statistical and neural models is highly context-dependent, with neural approaches not consistently outperforming classical methods in data-constrained settings (Spiliotis et al., 2022). This further motivates a careful empirical comparison rather than an a priori assumption of superiority for either approach.

The central problem, therefore, is to determine whether attention-based neural approaches can achieve predictive performance comparable to established Bayesian MMMs while providing additional structural insight into temporal and cross-channel marketing dynamics under realistic industrial constraints, without sacrificing robustness or usability.

1.2.1 Problem definition in the industrial context

The original problem motivating this thesis is rooted in an industrial need to improve the measurement of marketing effectiveness using aggregated data. At the host organization, MMM is employed as a core analytical tool to inform strategic marketing decisions across multiple brands and markets. The existing production pipeline is based on a Bayesian MMM framework that has proven reliable and interpretable in practice.

Despite its strengths, the current approach faces challenges when applied to increasingly complex marketing ecosystems. In particular, the growing

number of channels, the presence of strong correlations between paid and organic signals, and the emergence of long-term and indirect effects complicate the attribution of observed outcomes. These challenges raise practical questions regarding the adequacy of fixed functional forms and time-invariant assumptions commonly used in Bayesian MMMs (Chan and Perry, 2017; Ng et al., 2021).

From an industrial perspective, the problem is therefore defined as assessing whether alternative modeling approaches can capture the dynamics of modern marketing systems while remaining compatible with the operational constraints and decision-making needs of practitioners.

1.2.2 Scientific and engineering issues

Beyond the specific industrial setting, this thesis addresses a broader scientific and engineering question concerning the modeling of aggregated marketing systems. From an engineering perspective, the key issue is whether increased model expressiveness, in the form of attention-based neural architectures, can provide useful representations of non-linear, cross-channel, and long-term effects when data availability is limited. The specific research question derived from this problem discussion is formulated in Section 1.3.

1.3 Research Question

Based on the limitations identified in the problem discussion, this thesis addresses the following research question:

> **RQ:** To what extent can attention-based neural Marketing Mix Models achieve predictive performance comparable to Bayesian MMMs while enabling richer structural analyses of temporal and cross-channel marketing dynamics under realistic data constraints?

In the context of this thesis, realistic data constraints refer to the conditions typical of industrial marketing measurement: limited temporal coverage (approximately 140–240 weekly observations per merchant), high channel dimensionality (9–14 paid media channels), absence of user-level tracking or rich embedding signals, and high noise-to-signal ratios inherent to aggregated marketing data. While Mulc et al. (Mulc et al., 2025) demonstrate promising results for attention-based neural architectures in a setting with two channels

and rich embedding data, they explicitly identify generalization to more complex, data-constrained environments as an open question. Similarly, the broader time-series forecasting literature suggests that the relative advantage of neural models over statistical methods is highly context-dependent (Spiliotis et al., 2022), further motivating an empirical investigation under these specific conditions.

To answer this question, the thesis relies on two complementary evaluation dimensions. The first is *predictive comparability*, assessing whether the neural model can reach a level of out-of-sample forecasting performance comparable to a production-grade Bayesian MMM baseline, using R^2 and WMAPE. The second is *structural analysis*, examining whether the neural model enables richer inspection of temporal response profiles and cross-channel interaction patterns than models based on predefined parametric transformations and independent channel effects.

Addressing this question contributes to the understanding of how modern sequence models behave in low-sample, high-noise settings, and whether learned representations can provide meaningful analytical value when granular data are unavailable. The thesis approaches this through an empirical comparison between a production-grade Bayesian MMM baseline and an attention-based neural architecture inspired by the NNN framework (Mulc et al., 2025), evaluated on real-world aggregated marketing data from three e-commerce merchants.

Ethical considerations are addressed by relying exclusively on aggregated data, avoiding the use of user-level information and minimizing privacy risks. Sustainability considerations primarily concern computational efficiency: model complexity and experimentation are deliberately constrained to avoid unnecessary resource consumption, and the analysis emphasizes methodological rigor over brute-force optimization.

1.4 Goals

The thesis is structured around the following sub-goals:

1. **Scientific and engineering contribution.** Assess whether attention-based neural architectures can achieve predictive performance comparable to Bayesian MMMs while enabling richer analysis of temporal effects, cross-channel interactions, and long-term dependencies in low-sample, high-noise settings, contributing empirical insights to the literature on marketing measurement and sequence modeling.

2. **Industrial relevance.** Compare the predictive performance and structural modeling capabilities of attention-based neural models and a production-grade Bayesian MMM using real-world aggregated marketing data.

3. **Educational objectives.** Demonstrate mastery of advanced modeling techniques, experimental design, and critical evaluation by implementing, validating, and comparing multiple modeling approaches in a reproducible and methodologically sound manner.

The primary deliverables of the project include a reproducible experimental pipeline, a structured comparative analysis of modeling approaches, and a written thesis documenting the methodology, results, limitations, and implications of the study.

Beyond its applied goals, this thesis makes three scientific contributions. First, it provides empirical evidence that attention-based architectures, when applied under realistic data constraints, learn temporal response profiles that are substantially more heterogeneous than those obtainable through parametric adstock functions, demonstrating that the temporal structure of marketing effects is richer than what classical functional forms can represent. Second, the cross-channel interaction module introduced in this work quantifies inter-channel spillover effects from observational data without pre-specified functional forms, producing patterns consistent with established marketing funnel theory. Third, the analysis reveals a lag-zero overfitting tendency that highlights a fundamental trade-off between model flexibility and inductive bias in low-sample settings, contributing to the broader understanding of deep learning applied to small-sample time series.

1.5 Research Methodology

This thesis adopts an empirical and comparative research methodology. A production-grade Bayesian Marketing Mix Model serves as the baseline and is compared with attention-based neural models using aggregated time-series data from real-world industrial settings. Predictive performance is evaluated on out-of-sample observations using temporal data splits, complemented by diagnostic analyses of structural model behavior with respect to temporal dynamics and cross-channel interactions. The detailed discussion and motivation of the methodological choices are presented in Chapter 3.

1.6 Delimitations

The research question implies a broad investigation of attention-based neural models and Bayesian MMMs; however, the following aspects are deliberately excluded from the scope of this thesis.

First, the scope of the neural modeling approach is intentionally constrained. Not all architectural extensions proposed in the literature, such as cross-geographical attention mechanisms or advanced embedding aggregation strategies, are implemented. These extensions are acknowledged as promising directions for future research but are excluded to limit model complexity and reduce the risk of overfitting in data-constrained settings (Mulc et al., 2025).

Second, the comparison on the Bayesian side is limited to a single production-grade MMM baseline. Alternative Bayesian formulations or recent extensions, such as time-varying coefficient models, are discussed in the literature review but are not included as additional baselines in the empirical evaluation.

Third, the empirical analysis is conducted exclusively on data from e-commerce merchants. The findings may not generalize to other industries or domains where marketing dynamics, data availability, or channel structures differ substantially.

1.7 Structure of the thesis

The remainder of this thesis is structured as follows. Chapter 2 provides a detailed technical background and reviews related work on Marketing Mix Modeling, Bayesian approaches, and neural network–based methods for marketing measurement. Chapter 3 describes the data, modeling approaches, and experimental methodology used in the study.

Chapter 4 presents the implementation details of the models and the experimental setup. Chapter 5 reports the empirical results of the comparative evaluation, while Chapter 6 discusses the findings in relation to the research question and existing literature. Finally, Chapter 7 summarizes the contributions of the thesis and outlines directions for future research.

Chapter 2

Background

This chapter provides the technical background and literature review necessary to contextualize the research question addressed in this thesis. It introduces the core concepts of Marketing Mix Modeling (MMM), reviews established Bayesian approaches and their limitations, and discusses recent methodological extensions proposed in the literature. The chapter also positions neural and attention-based approaches within the broader landscape of marketing measurement research.

The chapter has two purposes. First, it equips the reader with the conceptual and methodological foundations required to understand the modeling choices made in later chapters. Second, it situates the present work within existing research, clarifying how prior studies are applied, extended, or deliberately not adopted in this thesis.

2.1 Marketing Mix Model

Marketing Mix Modeling (MMM) is a class of analytical techniques used to estimate the contribution of marketing activities to business outcomes such as sales or revenue, using aggregated time-series data. MMM has been widely adopted in industry due to its ability to operate without user-level tracking, making it suitable in contexts constrained by privacy regulations, limited experimentation capabilities, or fragmented data infrastructures.

At a high level, MMM models the relationship between a response variable and a set of marketing inputs, typically including media channels, promotions, and control variables such as seasonality, trends, and external factors. The objective is to infer how changes in marketing activity are associated with changes in the observed outcome over time. Because the data

are observational and aggregated, MMM relies on structural assumptions to achieve identifiability and statistical robustness.

Classical MMM implementations are commonly based on linear or generalized linear regression frameworks, augmented with domain-specific transformations designed to capture key characteristics of marketing effects. Among these, carryover effects are modeled using adstock functions, while diminishing returns are represented through saturation or response curves (Chan and Perry, 2017; Broadbent, 1979). These transformations encode prior assumptions about consumer response dynamics and play a central role in shaping model behavior. Bayesian formulations of MMM have become particularly prevalent in practice. By incorporating prior distributions over model parameters, Bayesian MMMs offer a principled way to regularize estimates, handle multicollinearity among channels, and quantify uncertainty in inferred effects. This makes them attractive for decision support, where interpretability and stability are often prioritized over raw predictive accuracy Chan and Perry (2017).

2.1.1 Bayesian Marketing Mix Models

Bayesian Marketing Mix Models extend classical regression-based MMMs by explicitly modeling uncertainty and incorporating prior knowledge about channel effects. Priors are commonly used to enforce sign constraints, shrink coefficients toward plausible ranges, and stabilize estimation in settings with limited data. This is particularly important in MMM, where the number of parameters can be large relative to the number of time points, and where marketing channels are often highly correlated (Chan and Perry, 2017).

2.1.1.1 Model Specification

A Bayesian MMM defines a generative model for the observed target y_t at each time step t. The standard formulation decomposes the expected response into an intercept (or baseline), a sum of channel contributions, control effects, and a seasonality component:

$$y_t \sim \mathcal{N}(\mu_t, \sigma^2), \qquad \mu_t = b_t + \sum_{c=1}^{C} m_{t,c} + \sum_j \gamma_j z_{t,j} + s_t\,, \tag{2.1}$$

where b_t is a baseline term capturing organic demand, $m_{t,c}$ is the contribution of media channel c, γ_j are coefficients on control variables $z_{t,j}$ (e.g. lagged targets, macroeconomic indicators), and s_t represents seasonal effects

typically modeled via Fourier basis functions. The observation noise σ is assigned a weakly informative prior (e.g. $\sigma \sim \text{HalfNormal}$).

Each channel contribution $m_{t,c}$ is obtained by applying two sequential transformations to the raw media spend $x_{t,c}$. The first is an *adstock* (carryover) function that distributes the effect of spend across subsequent time periods. For example, under a Weibull parameterization (Weibull, 1951; Broadbent, 1979; Jin et al., 2017), the adstocked spend is a convolution of the raw spend with a normalized kernel:

$$\tilde{x}_{t,c} = \sum_{\ell=0}^{L} w_\ell(k_c, \lambda_c)\, x_{t-\ell,c}, \qquad w_\ell \propto f_{\text{Weibull}}(\ell;\, k_c, \lambda_c)\,, \tag{2.2}$$

where k_c and λ_c are per-channel shape and scale parameters controlling the decay profile, and L is a maximum lag. The second transformation is a *saturation* function that models diminishing returns. A common choice is the Michaelis–Menten form (Michaelis and Menten, 1913):

$$m_{t,c} = \frac{\beta_c\, \tilde{x}_{t,c}}{\beta_c\, \rho_c + \tilde{x}_{t,c}}\,, \tag{2.3}$$

where $\beta_c > 0$ controls the asymptotic contribution and $\rho_c > 0$ determines the half-saturation point. All channel-level parameters (β_c, ρ_c, k_c, λ_c) are assigned independent prior distributions (e.g. Gamma or Beta), encoding domain knowledge about plausible response magnitudes and temporal decay rates.

2.1.1.2 Inference

Posterior inference is performed via Markov Chain Monte Carlo (MCMC), typically using the No-U-Turn Sampler (NUTS) (Hoffman and Gelman, 2014; Chan and Perry, 2017). NUTS is a gradient-based sampler that automatically tunes step size and trajectory length, making it well suited for the moderate-dimensional parameter spaces encountered in MMM. The posterior predictive distribution is obtained by propagating posterior samples through the generative model; point predictions are typically reported as the posterior mean, accompanied by credible intervals to quantify uncertainty.

2.1.1.3 Limitations

Despite their practical advantages, Bayesian MMMs exhibit several limitations. Their expressiveness is constrained by the choice of parametric

transformations: adstock and saturation functions must be specified a priori, and their functional forms may not adequately capture complex or evolving response patterns (Chan and Perry, 2017). Channel contributions enter additively with no interaction terms, so cross-channel synergies and mediated effects are not represented; treating channels as independent regressors can lead to biased estimates when the underlying causal structure involves inter-channel dependencies (Gong et al., 2024). Finally, many Bayesian MMMs assume time-invariant parameters. While extensions such as time-varying coefficient models relax this assumption (Ng et al., 2021), they increase complexity while remaining dependent on handcrafted functional forms.

2.2 Neural and Attention-Based Approaches to Marketing Mix Modeling

Recent advances in deep learning (Goodfellow et al., 2016) for sequential data have motivated the exploration of neural architectures as alternatives to traditional regression-based Marketing Mix Models. In particular, attention-based models have shown strong performance in settings characterized by long-term dependencies, complex interactions, and heterogeneous inputs, making them a promising candidate for marketing measurement tasks (Vaswani et al., 2017; Lim et al., 2021). Early neural approaches to MMM primarily focused on predictive accuracy, often at the expense of interpretability and stability. More recent work has sought to incorporate structural inductive biases and domain-specific constraints, aiming to retain the strengths of classical MMM while extending their expressive capacity.

A prominent example of this direction is the Neural Next-generation Networks (NNN) framework proposed by Google (Mulc et al., 2025). NNN replaces the scalar channel representations and parametric transformations of Bayesian MMMs with embedding-based inputs and learned attention mechanisms. The remainder of this section describes the NNN architecture formally; Chapter 4 then details how this thesis adapts and extends it.

2.2.1 The NNN Framework

2.2.1.1 Input Representation

NNN operates on a rank-4 tensor $X \in \mathbb{R}^{G \times T \times C \times D}$, where G is the number of geographic regions, T the number of time steps, C the total number of channels (media, organic, and target), and D the embedding dimension. Each channel

entry $\mathbf{x}_{g,t,c} \in \mathbb{R}^D$ encodes both the *volume* of activity (through its magnitude $\|\mathbf{x}_{g,t,c}\|$) and its *qualitative content* (through its direction). Embeddings are obtained from pretrained models: a language model for search queries and a video model for ad creatives. For channels represented only as scalars (e.g. spend), the scalar is placed in the first coordinate and the remaining $D - 1$ dimensions are zero-padded, preserving the magnitude convention.

2.2.1.2 Transformer Architecture

The input tensor is processed by N identical transformer layers, each consisting of a factored self-attention block followed by a channel-wise MLP, with residual connections (He et al., 2016) after both operations:

$$X^{(n)} = \text{Transformer}(X^{(n-1)}, \Theta), \qquad n = 1, \dots, N, \tag{2.4}$$

with $X^{(0)} = X$. Unlike standard transformers (Vaswani et al., 2017), NNN omits layer normalization because the vector magnitudes carry volume information that must be preserved.

2.2.1.3 Factored Self-Attention

Instead of computing a single $TC \times TC$ attention matrix, NNN factorizes attention into two independent components.

Temporal attention. An MLP $a_{\text{time}} : \mathbb{R}^{1+C} \to \mathbb{R}$ maps the normalized time lag $\Delta_{t,t'} = (t - t')/\omega$ and a one-hot channel identifier $E_c \in \mathbb{R}^C$ to a raw score $s_{t,t',c} = a_{\text{time}}(\Delta_{t,t'}, E_c)$. A causal mask zeroes out future positions ($t' > t$) and entries beyond the lookback window ω. Temporal attention weights are obtained via scaled softmax:

$$(W_{\text{time}})_{t,c,t'} = \frac{\exp(s_{t,t',c} \,/\, \tau)}{\sum_{t''} \exp(s_{t,t'',c} \,/\, \tau)}, \tag{2.5}$$

where τ is a temperature parameter.

Channel attention. A learnable matrix $\psi \in \mathbb{R}^{C \times C}$, initialized to the identity, captures cross-channel affinity. Channel attention weights are computed by applying softmax along the source channel dimension: $(W_{\text{chan}})_{c,c'} = \text{softmax}_{c'}(\psi_{c,:} \,/\, \tau)$.

Factored output. The two components are combined multiplicatively and applied to the input:

$$O_{g,t,c,d} = \sum_{t'} \sum_{c'} (W_{\text{time}})_{t,c,t'} \cdot (W_{\text{chan}})_{c,c'} \cdot X_{g,t',c',d} \,. \tag{2.6}$$

This factorization reduces the attention complexity from $O((TC)^2)$ to $O(T^2 + C^2)$ and provides an interpretable decomposition into *when* and *from which channel* information flows.

2.2.1.4 Prediction Heads and Training

After the transformer layers, separate prediction heads produce the model outputs. The *sales head* combines channel volumes with a learned intent function:

$$\hat{y}_{g,t} = \sum_{c} V_{g,t,c} \cdot P\big(\bar{X}^{(N)}_{g,t,c},\, g,\, \theta_c\big)\,, \tag{2.7}$$

where $V_{g,t,c} = \|X^{(N)}_{g,t,c}\|$ is the channel volume, $\bar{X}^{(N)}_{g,t,c}$ the unit-normalized direction, and P is an MLP with sigmoid output that acts as a learned saturation function. A *search head* predicts the next-step search embedding via a separate MLP, enabling multitask learning. The model is trained by minimizing a combined loss with L1 regularization:

$$\mathcal{L} = \alpha\, \mathcal{L}_{\text{sales}} + (1 - \alpha)\, \mathcal{L}_{\text{search}} + \lambda \sum_{\theta \in \Theta} |\theta| \,. \tag{2.8}$$

2.2.1.5 Counterfactual Attribution

Channel attribution is computed via counterfactual analysis. The contribution of channel c is defined as the difference in predicted sales between the original input and a modified input $\tilde{X}_c$ in which all activity from channel c is set to zero:

$$\text{Attribution}_c = \sum_{g,t} \hat{y}(X)_{g,t} - \hat{y}(\tilde{X}_c)_{g,t} \,. \tag{2.9}$$

Results are typically reported as a mix percentage,

$$\text{mix\%}_c = \frac{\text{Attribution}_c}{\sum_{c'} \text{Attribution}_{c'}} \,. \tag{2.10}$$

When the model includes a search prediction head, autoregressive unrolling can propagate the removal of a channel through the intermediary signals,

capturing indirect (mediated) effects.

2.2.1.6 Scope and Limitations

NNN introduces assumptions and practical requirements. The framework relies on the availability of rich embedding signals and intermediary channels, and its empirical validation has focused on settings with only two paid media channels. The authors note that multi-head attention could allow different heads to specialize in distinct temporal patterns. As a result, NNN represents a compelling but non-trivial starting point whose applicability to data-constrained, multi-channel industrial settings remains to be established.

This thesis builds on the ideas introduced by NNN, with a specific focus on evaluating their applicability under realistic industrial constraints. In particular, it investigates whether attention-based architectures can provide measurable benefits when applied to aggregated marketing data with a high number of channels and limited embedding signals, and how their performance compares to a Bayesian MMM baseline. The specific architectural adaptations are described in Chapter 4.

2.3 Related Work

This section situates the present work within the broader research landscape by reviewing contributions along the main methodological references the thesis builds upon: Bayesian and classical MMM, causal extensions, attention-based time-series modeling, embedding methods for structured data, multitask learning, neural approaches to marketing measurement, and simulation frameworks for model evaluation.

2.3.1 Classical and Bayesian Extensions of Marketing Mix Modeling

The econometric foundations of Marketing Mix Modeling date back to market response models that formalize the relationship between marketing inputs and sales outcomes (Hanssens et al., 2001). For a recent survey of the evolution from classical regression to modern AI-powered approaches, see Gujar et al. (Gujar et al., 2024). Modern Bayesian formulations extend this line by incorporating prior distributions to regularize estimates, handle multicollinearity, and quantify uncertainty, making them the dominant paradigm in industrial applications (Chan and Perry, 2017). Jin et al. further

develop Bayesian MMMs by explicitly modeling carryover and shape effects through flexible parameterizations of adstock and saturation curves (Jin et al., 2017), while Zhang et al. propose calibrating these models with ROI priors derived from lift test experiments to improve the reliability of channel-level estimates (Zhang et al., 2024). On the frequentist side, Runge et al. introduce Robyn, a semi-automated open-source framework that replaces MCMC inference with gradient-based optimization and multi-objective model selection (Runge et al., 2024). It is worth noting that much of the foundational work on Marketing Mix Modeling has originated from industry research groups and has been disseminated as technical reports or preprints rather than through traditional peer-reviewed venues, reflecting both the applied nature of the field and proprietary constraints on data sharing.

Despite these advances, Bayesian MMMs remain constrained by strong parametric assumptions. Channel effects require predefined functional forms, and cross-channel interactions are typically either ignored or handled through ad-hoc interaction terms (Naik and Raman, 2003). Time-varying coefficient models relax the stationarity assumption (Ng et al., 2021), but increase complexity while remaining dependent on handcrafted specifications. Dew et al. demonstrate a more fundamental issue: nonlinear and time-varying effects are often not separately identifiable when marketing variables are autocorrelated, and conflating the two can lead to suboptimal budget allocations (Dew et al., 2024). As a result, Bayesian MMMs provide a strong and interpretable baseline but face limitations in capturing complex, long-term, and mediated marketing dynamics.

2.3.2 Causal and Structural Approaches to MMM

Recent work has explored causal formulations of Marketing Mix Modeling to address limitations of purely associational approaches. Gong et al. (Gong et al., 2024) propose CausalMMM, which integrates causal discovery with MMM to infer directed relationships among marketing variables. While causal approaches offer conceptual advantages, they rely on strong assumptions about the data-generating process and are sensitive to model misspecification. Scalability and robustness under noisy, aggregated data conditions remain open challenges.

2.3.3 Attention Mechanisms and Transformers for Time Series

The Transformer architecture (Vaswani et al., 2017) introduced scaled dot-product self-attention as a general mechanism for modeling dependencies in sequential data, replacing recurrence with parallelizable pairwise interactions. Its success in natural language processing motivated adaptations to time-series forecasting, where long-range temporal dependencies and multivariate interactions pose similar challenges.

Lim et al. propose Temporal Fusion Transformers (TFT), which combine multi-head attention with variable selection networks and interpretable temporal gates, achieving strong forecasting performance while retaining explainability (Lim et al., 2021). Zhou et al. address the quadratic cost of full attention through ProbSparse self-attention in the Informer architecture, reducing complexity to $O(L \log L)$ and enabling efficient long-sequence forecasting (Zhou et al., 2021).

These developments are directly relevant to the present work: the multi-scale temporal attention mechanism adopted in this thesis draws on the same principle of learning temporal dependencies through attention weights, while introducing domain-specific factorizations (temporal vs. channel) and multi-resolution pooling tailored to marketing data.

2.3.4 Embedding Methods for Structured Data

Representing categorical variables as dense, low-dimensional vectors learned jointly with the task objective has become standard practice in deep learning for tabular and structured data. Guo and Berkhahn show that entity embeddings of categorical variables outperform one-hot encoding and reveal meaningful latent structure, with embeddings of similar categories clustering together in the learned space (Guo and Berkhahn, 2016). This thesis applies the same principle to marketing channels, constructing hierarchical embeddings that capture group, channel, and funnel-stage structure.

2.3.5 Multitask Learning

Multitask learning improves generalization by sharing representations across related objectives, providing an inductive bias that is particularly valuable under limited data (Caruana, 1997). In the marketing measurement context, jointly predicting a primary business outcome and an auxiliary marketing indicator (e.g. impressions) offers a denser supervision signal that stabilizes

representation learning. This strategy is adopted by the NNN framework and extended in the present work.

2.3.6 Neural Approaches to Marketing Measurement

Neural approaches to marketing measurement have been investigated as a means to overcome the rigidity of traditional MMMs. The Neural Next-generation Networks (NNN) framework represents the most structured effort in this direction, combining embedding-based representations, factored self-attention, and multitask learning with intermediary signals (Mulc et al., 2025). Complementary work on neural temporal point processes (Mei and Eisner, 2017) provides a related formalism for modeling event-driven interactions across channels, though it targets different data regimes.

While NNN shows promising results in modeling long-term and mediated effects, its evaluation has focused on data-rich settings with only two paid media channels. The applicability of attention-based architectures to aggregated industrial datasets with many channels and limited auxiliary signals remains an open question that this thesis directly addresses.

2.3.7 Simulation and Evaluation Frameworks

Zhang and Vaver propose the Aggregate Marketing System Simulator (AMSS), a simulation framework for benchmarking MMM methods under controlled conditions with known ground-truth effects (Vaver and Zhang, 2017). Such frameworks improve experimental rigor and comparability across models, but do not address the expressiveness limitations of existing MMM architectures.

Chapter 3

Methodology

This chapter describes the research methodology adopted to address the research question introduced in Section 1.3. The chapter first discusses the choice of research method, including the rationale for selecting an empirical comparative approach over alternative methodologies. It then describes in detail how the chosen method is applied, covering data collection, experimental design, and the technical environment. Finally, the chapter discusses reliability, validity, and the evaluation framework used to analyze the results.

3.1 Choice of research method

The research question addressed in this thesis asks to what extent attention-based neural Marketing Mix Models can achieve predictive performance comparable to Bayesian Marketing Mix Models while enabling richer structural analyses under realistic data constraints. Answering this question requires a methodology that enables controlled comparison of modeling approaches on real-world data, with measurable and reproducible outcomes.

3.1.1 Research paradigm

The thesis adopts a primarily positivist research paradigm, in which knowledge is generated through empirical observation and quantitative evaluation of observable phenomena. Competing modeling paradigms are evaluated using observational time-series data and objective performance metrics, allowing the analysis to focus on measurable model behavior rather than on subjective interpretation. At the same time, the work reflects a pragmatic engineering perspective: the analysis is conducted using real-world industrial data and an

existing production modeling pipeline, enabling the assessment of practical applicability under realistic operational constraints.

Within this framework, conclusions are drawn through two complementary forms of analysis. Predictive evaluation metrics are used to compare the accuracy of the competing models, while diagnostic analyses investigate how each approach represents temporal dynamics and cross-channel interactions. This distinction reflects the dual nature of the research question, where predictive performance can be objectively measured, while structural insights require interpretation in the absence of ground truth.

3.1.2 Alternative methodologies considered

Several alternative research methodologies were considered for addressing the research question.

Controlled experiments. Randomized online experiments and causal interventions would provide the strongest evidence for assessing model validity. However, such experiments require the ability to manipulate marketing activity in a live environment, which is outside the scope and control of this thesis due to practical and organizational constraints.

Simulation-based studies. Synthetic data generated from known data-generating processes would allow evaluation against ground-truth effects. While useful for benchmarking, purely simulation-based studies do not assess model behavior on real-world data, where noise, confounding, and distributional characteristics differ from synthetic settings. The primary objective of this thesis is to evaluate model performance and structural behavior under realistic industrial conditions, making a simulation-only approach insufficient.

Qualitative and survey-based methods. Approaches based on expert interviews or practitioner surveys could provide complementary insights into model usability and adoption. However, the research question is inherently quantitative and model-driven, requiring objective performance evaluation rather than subjective assessment.

Purely theoretical analysis. A formal mathematical comparison of model classes could characterize differences in expressiveness and approximation capacity. However, such an analysis would not address the empirical question of how the models perform under the specific data constraints encountered in practice.

3.1.3 Rationale for the chosen method

Based on this analysis, an empirical and comparative research methodology is adopted. This approach directly addresses the research question by implementing and evaluating both modeling paradigms on the same real-world datasets under identical conditions. The comparison is conducted using aggregated time-series data from industrial settings, reflecting the practical constraints under which Marketing Mix Models are typically deployed.

This methodology is grounded in a quantitative evaluation strategy relying on time-based train–validation–test splits to assess out-of-sample predictive performance while avoiding temporal leakage. Standard predictive error metrics are complemented by targeted diagnostic analyses designed to probe structural model behavior with respect to temporal dynamics and cross-channel interactions.

3.2 Application of research method

This section describes in detail how the chosen empirical comparative methodology is applied, covering the research process, data collection, experimental design, and the technical environment. The level of detail is intended to ensure that the study can be reproduced.

3.2.1 Research process

The research process combines elements of academic investigation and engineering development, and can be summarized in five stages: (1) problem formulation, identifying limitations in current MMM approaches; (2) literature review, establishing theoretical foundations across Bayesian inference and neural sequence modeling; (3) model development, implementing an attention-based neural architecture for aggregated marketing data; (4) experimental evaluation, assessing both models using a consistent protocol and identical datasets; and (5) comparative analysis, analyzing differences in predictive performance and modeling behavior to address the research question.

3.2.2 Data collection

The empirical analysis in this thesis relies on aggregated marketing and business data obtained from the analytics infrastructure of the host organization. The dataset consists of historical observations of marketing

activity and business outcomes collected from several e-commerce merchants operating across multiple markets.

The data include weekly measurements of marketing spend across a set of advertising channels, together with business performance indicators such as sales or profitability metrics. These observations form multivariate time series describing the evolution of marketing activity and outcomes over time. The dataset reflects the type of aggregated information typically available in industrial Marketing Mix Modeling applications, where individual user-level data are not accessible due to privacy, regulatory, or infrastructure constraints.

Sampling

The data used in this study are obtained from a limited number of merchants operating on the same analytics platform. For each merchant, historical marketing and sales data are collected across multiple channels and markets. The analysis focuses on aggregated weekly observations, which represent the standard temporal granularity used in Marketing Mix Modeling.

Sample Size

The dataset covers approximately three to four years of historical observations, corresponding to roughly 200 weekly data points per merchant. This is a relatively limited sample size, especially for expressive neural models, and some minor marketing channels are characterized by sparse or irregular spending patterns, resulting in low signal density. At the same time, this time horizon is consistent with real-world MMM practice: using substantially longer histories is not necessarily desirable, since e-commerce businesses often undergo changes in market conditions, channel mix, customer behavior, and commercial strategy over relatively short periods of time. As a result, the model must operate on a moderate amount of aggregated data that is recent enough to remain relevant, even though such data may not always be sufficient to reliably identify complex or long-term response dynamics.

Target Population

The target population of this study consists of e-commerce businesses that rely on aggregated marketing data to evaluate the effectiveness of advertising investments. Such organizations commonly operate multiple marketing channels and geographic markets while lacking access to user-level attribution

data. The dataset used in this thesis therefore reflects a realistic industrial setting in which Marketing Mix Models are typically applied.

3.2.3 Experimental design and planned measurements

This subsection describes the experimental setup used to evaluate the neural marketing model and compare its behavior with an existing industrial Marketing Mix Modeling (MMM) baseline. The experiments are conducted on aggregated marketing time series derived from real-world e-commerce data, comparing the predictive performance of the neural model with the outputs produced by the existing MMM pipeline while also enabling diagnostic analyses on the internal structure learned by the neural architecture.

3.2.3.1 Test environment and model

The experiments are conducted in an offline research environment using historical marketing data obtained from the organization's analytics infrastructure. The dataset consists of aggregated time series describing marketing activity and business outcomes for several e-commerce merchants.

Each observation corresponds to a weekly time step and includes marketing spend across multiple advertising channels together with a business performance indicator used as the prediction target. In this work the primary target variable is Net Gross Profit, which represents the economic value of sales after accounting for returns and discounts. The neural model operates on multivariate time series derived from the aggregated dataset, predicting the weekly value of the target metric. In addition to the primary target, the model optionally predicts an intermediate signal related to advertising exposure in a multitask learning setting, which helps stabilize the representation of marketing activity during training (Mulc et al., 2025).

The neural architecture is designed to capture both temporal dynamics and interactions between marketing channels. Temporal effects are modeled through a multi-scale attention mechanism, in which multiple temporal windows of different lengths are considered in parallel. Each temporal scale focuses on a specific time horizon, ranging from short-term effects to longer-term delayed responses. For each scale, the model applies channel-specific temporal weighting functions over past observations within the corresponding lookback window, and the outputs are combined through learned weights. This design enables the model to represent heterogeneous

temporal dynamics across marketing channels, playing a role analogous to adstock transformations but without imposing a fixed parametric form.

Interactions between channels are modeled through temporal cross-channel attention mechanisms, with cross-channel effects represented through channel-pair–specific impulse response functions (IRFs) parameterized using Weibull kernels. This formulation allows the model to capture flexible, lagged relationships between channels under limited and aggregated data conditions. Traditional Bayesian MMM models typically do not explicitly model such cross-channel lagged effects, as their structure focuses primarily on independent channel contributions with predefined transformations.

Channel-level attribution is estimated using a counterfactual leave-one-out procedure. For each channel, the model prediction obtained with all inputs is compared with the prediction obtained when the channel is removed from the input. The difference between these predictions provides an estimate of the incremental contribution of the channel to the observed outcome.

3.2.3.2 Training protocol and baseline comparison

Model training follows a supervised learning setup using historical time series data. The dataset is split into training, validation, and test segments using a temporal holdout strategy in order to preserve the chronological order of observations. Early stopping and learning rate scheduling are used during training to stabilize convergence.

Predictive performance is evaluated using the coefficient of determination (R^2) and the Weighted Mean Absolute Percentage Error (WMAPE), which provide complementary information about variance explanation and prediction error relative to the scale of the business metric.

The comparison baseline is an existing Bayesian Marketing Mix Model implemented in the production analytics system. The MMM decomposes the target metric into contributions from marketing channels, baseline demand, and control variables. Because the two modeling paradigms rely on different structural assumptions, the comparison focuses on predictive performance and on the interpretation of model behavior rather than on establishing a single ground-truth attribution of marketing effects.

3.2.3.3 Hyperparameter selection

The neural model includes several hyperparameters controlling training stability, model capacity, and baseline structure. The search procedure

is organized into sequential phases that progressively refine the model configuration.

The first phase focuses on parameters affecting training stability, such as the learning rate and regularization strength. Once a stable training regime is identified, the second phase explores parameters related to model capacity, including the dimensionality of internal representations.

In a final phase the configuration of the baseline component is refined. This component models non-marketing demand signals and interacts with the media contribution estimates produced by the network.

Hyperparameter configurations are evaluated using a composite score based on predictive performance metrics (R^2 and WMAPE), ensuring that selected models both explain variance in the target variable and maintain low prediction errors relative to the overall scale of the business metric.

3.2.4 Hardware and software environment

All experiments were conducted in a Python-based research environment designed for machine learning experimentation. The neural model was implemented using the PyTorch deep learning framework (Paszke et al., 2019). Data preprocessing and experiment orchestration were implemented using pandas and NumPy.

Model configuration and experiment settings were managed through structured configuration files, allowing different merchants and datasets to be processed using consistent training procedures. The experimental workflow consists of data preparation, model training, hyperparameter search, and post-training analysis. All experiments were executed in an offline environment using historical data, ensuring that no live production systems were affected during the research process.

The codebase follows standard software engineering practices to ensure reproducibility and maintainability, including dependency management through a controlled Python environment and automated code quality checks.

3.3 Assessing reliability and validity of the data collected

This section describes the methodological measures adopted to ensure the validity and reliability of the experimental results.

3.3.1 Validity of method

Both models are evaluated on the same aggregated marketing dataset and using the same business target variable. Performance metrics are computed on out-of-sample observations obtained through temporal data splitting, ensuring that predictions are evaluated on future time periods not used during model training.

The comparison focuses on predictive performance and diagnostic analyses derived directly from the model outputs. Because the dataset is observational, the analysis does not attempt to establish causal identification and instead evaluates model behavior under the available data constraints.

3.3.2 Reliability of method

Several measures are used to increase the reliability of the modeling procedure. Model training uses fixed random seeds to reduce variability introduced by stochastic optimization. Early stopping based on validation performance is applied to stabilize training and reduce the risk of overfitting. Hyperparameter search follows a structured procedure to ensure that candidate configurations are evaluated using consistent criteria.

These measures help ensure that the reported model performance reflects stable patterns in the data rather than random fluctuations in the training process.

3.3.3 Data validity

The dataset used in this study is derived from the organization's internal analytics infrastructure and contains aggregated marketing and sales data from real e-commerce operations. Observations are aggregated at weekly frequency and include marketing investments across multiple advertising channels together with a business performance metric used as the prediction target.

The use of aggregated data avoids the processing of user-level information while still capturing the overall dynamics of marketing activity and business outcomes.

3.3.4 Reliability of data

The dataset is generated through a deterministic data-processing pipeline that extracts marketing and sales information from the analytics infrastructure and transforms it into a structured time-series representation. Because the

pipeline follows a fixed processing procedure, the same input data produce identical datasets when the pipeline is executed again, ensuring reproducibility of the experiments.

3.4 Evaluation framework

The evaluation framework is designed to assess whether neural architectures can achieve predictive performance comparable to traditional Marketing Mix Modeling approaches while enabling richer structural analyses when applied to aggregated marketing data. The evaluation is structured around the research question introduced in Section 1.3 and focuses on two complementary dimensions.

3.4.1 Predictive performance

Predictive performance is evaluated to address the predictive comparability dimension of the research question, assessing whether the neural model captures the relationship between marketing activity and business outcomes at a level comparable to the Bayesian MMM baseline. Performance is measured on a holdout test set using two complementary metrics.

The coefficient of determination (R^2) measures the proportion of variance in the target variable explained by the model. The Weighted Mean Absolute Percentage Error (WMAPE) measures the total absolute prediction error relative to the total observed outcome over the test period, and is defined as:

$$\frac{\sum_{t=1}^{T} |y_t - \hat{y}t|}{\sum t = 1^{T} |y_t|} \times 100, . \tag{3.1}$$

where y_t denotes the observed target value, $\hat{y} * t$ the corresponding prediction, and $\widetilde{J} * $ test the set of test observations. Unlike standard MAPE, which averages percentage errors at each time step and can become unstable when observed values are small, WMAPE aggregates errors over the full test period before normalization. It can therefore be interpreted as the percentage of the total observed business outcome that is missed by the model in absolute terms; lower values indicate better predictive accuracy.

3.4.2 Model behavior and structural analysis

Model behavior is analyzed to address the structural analysis dimension of the research question, focusing on whether the neural model enables richer inspection of temporal dynamics and cross-channel interactions than the Bayesian MMM baseline. The analysis examines two dimensions: temporal effects, by inspecting the learned temporal response profiles across different time scales and comparing them with the parametric adstock mechanisms used in the Bayesian baseline; and cross-channel dynamics, by examining the interaction patterns learned between marketing channels.

Because no ground-truth decomposition of marketing effects is available, these structural analyses are evaluated pragmatically, focusing on their plausibility, stability, and usefulness for understanding the behavior of the marketing system rather than on strict correctness.

Detailed implementation aspects of the data preparation, model training, and analysis pipelines are described in Chapter 4.

Chapter 4

Implementation

This chapter describes the implementation of the neural system developed for the MMM task, focusing on the architectural design of the model, the data representation adopted for marketing time series, and the training strategy used during the empirical evaluation.

The objective of the system is to estimate the incremental contribution of marketing activities to observed economic outcomes.

The proposed architecture builds upon the NNN framework introduced in Mulc et al. (2025), but introduces several extensions motivated by specific limitations of the original experimental setting. The NNN paper validates its factored self-attention mechanism on a simplified scenario: two paid media channels, synthetic data with known ground-truth adstock parameters, and rich pretrained embeddings of dimension $D = 256$–512. In contrast, the real-world marketing datasets addressed in this thesis involve 9–14 marketing channels, scalar spend inputs projected to low-dimensional embedding vectors (representation vectors of dimension $D = 8$–32, detailed in Section 4.1), and approximately 140–240 weekly observations. Under these conditions, several design assumptions of the original architecture (notably the use of a single shared temporal MLP and a static cross-channel interaction matrix) become limiting, as discussed in the following sections.

A guiding principle throughout the design is *asymmetric model complexity*: the model allocates higher representational capacity where the learning signal is strong, and constrains capacity where the signal is weak. Within-channel temporal effects are modeled non-parametrically through per-channel attention functions, because this first-order relationship between a channel's past spending and its current contribution is strong enough to justify the flexibility. Cross-channel temporal effects, which are second-order

modulations attenuated through multiple layers, are modeled parametrically with a compact Weibull kernel requiring only two shape parameters per channel pair.

The chapter is organized as follows. First, the overall model architecture is presented, including the hierarchical channel embedding, the temporal attention mechanism, and the cross-channel interaction module. Next, the prediction structure and baseline decomposition are described. The final sections present the training objective, the optimization procedure, and the attribution mechanism used to interpret channel contributions.

4.1 Model Architecture

The proposed neural MMM system is organized as a sequential pipeline that mirrors the causal structure assumed in marketing response modeling. Raw media spend inputs are first projected into a shared latent space by a *hierarchical channel embedding* layer, which encodes structural relationships between marketing platforms, funnel stages, and channel groups. The resulting embeddings are then processed by a *factored attention* mechanism that decomposes temporal and cross-channel dynamics into two complementary components: per-channel temporal MLPs capturing heterogeneous carryover profiles, and a lag-dependent interaction kernel modeling delayed spillover effects between channels. Finally, the enriched representations are mapped to predictions through *per-channel sales heads* that produce non-negative media contributions, while a separate *baseline module* captures organic demand from lagged targets and calendar features. The predicted outcome is the sum of all channel effects and the scaled baseline.

Figure 4.1 illustrates the end-to-end data flow of the proposed architecture, from raw media spend inputs to the final prediction decomposed into channel-specific contributions and organic baseline.

4.1.1 Data Representation

The model operates on weekly aggregated marketing and business data collected across multiple merchants and marketing channels, as described in Section 3.2.2. Figure 4.2 summarizes the end-to-end data pipeline from the company database to the input tensor consumed by the model. Each temporal observation comprises the media spend for every active channel, a set of control variables representing organic demand signals (e.g., lagged

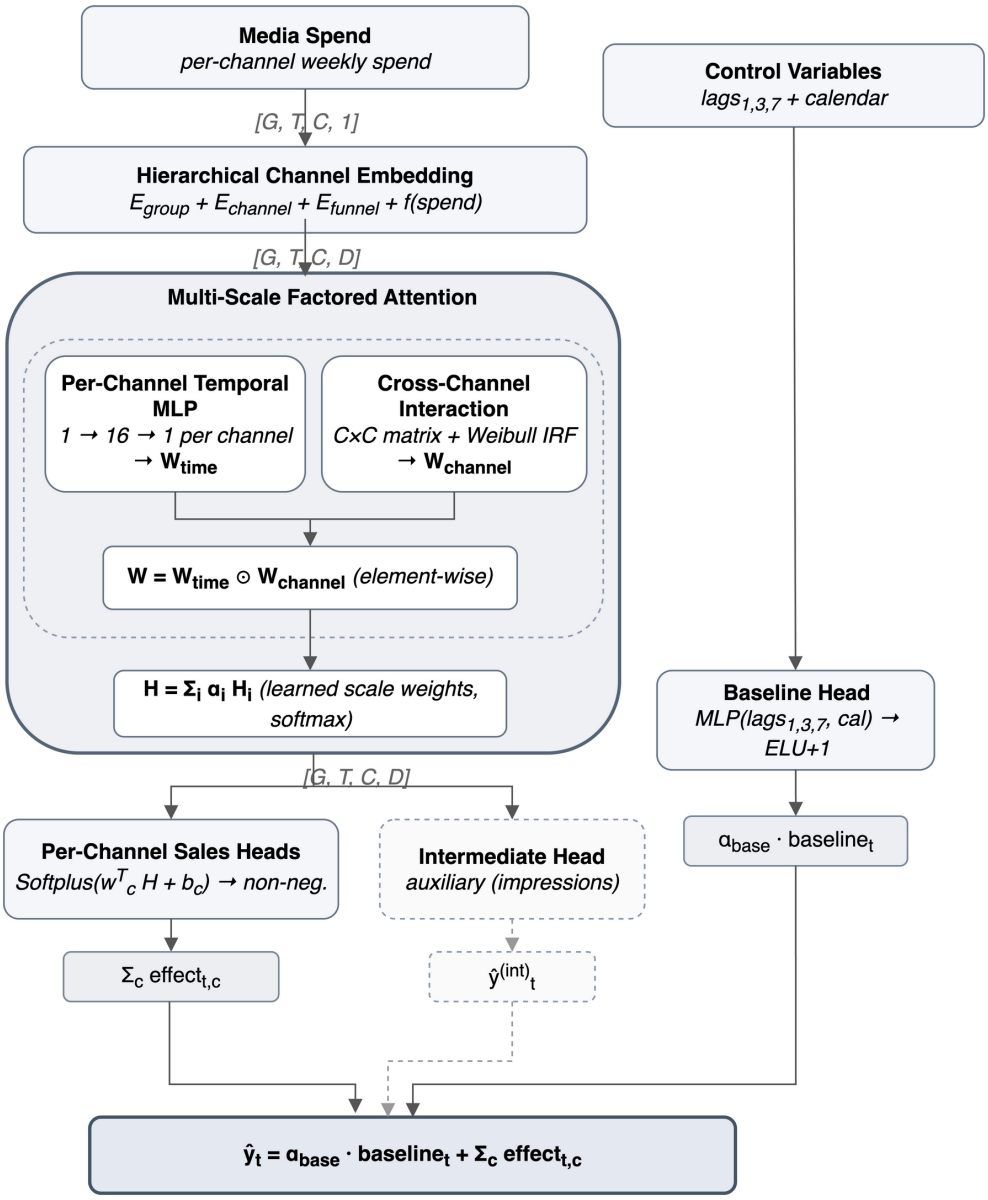


Figure 4.1: High-level architecture of the proposed neural Marketing Mix Modeling system. Media spend inputs are transformed through a hierarchical channel embedding and processed by a multi-scale factored attention mechanism with per-channel temporal MLPs and a lag-dependent cross-channel interaction kernel. Per-channel sales heads produce non-negative media contributions, while a separate baseline head captures organic demand. The final prediction is the sum of media effects and scaled baseline.

target values and calendar indicators), the primary economic target, and an intermediate marketing indicator used for multitask supervision.

Let $X \in \mathbb{R}^{G \times T \times C}$ denote the raw media spend tensor and $y \in \mathbb{R}^{T}$ the observed target series, where G, T, and C are the number of merchants, time steps, and channels respectively. Before training, all input features and target variables are normalized using a global scale factor derived from the maximum observed target:

$$s = \max\left(\frac{\max_t |y_t|}{50},\ 1\right), \qquad \tilde{X} = \frac{X}{s}, \qquad \tilde{y} = \frac{y}{s}. \tag{4.1}$$

The constant 50 maps the target maximum to a moderate numerical range compatible with standard Adam optimizer defaults (Kingma and Ba, 2015). This global linear scaling was chosen over log-transform, z-score standardization, and min-max scaling because it is the only option that preserves both cross-channel magnitude ratios and the shape of spend-response curves, properties that carry attribution-relevant information and that the alternatives distort. The intermediate target (impressions) is scaled independently with its own factor to account for its different magnitude. All subsequent model equations operate on the normalized quantities; the tilde is dropped for notational convenience.

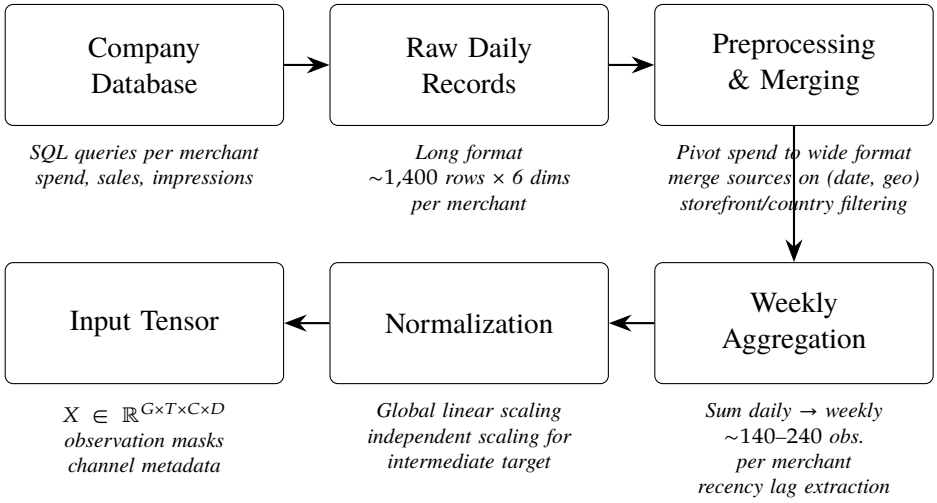


Figure 4.2: Data pipeline from the company database to the rank-4 input tensor consumed by the model. Raw daily records are fetched via SQL, preprocessed and merged into a single dataset per merchant, aggregated to weekly granularity, and normalized using a global scale factor.

4.1.1.1 Input Variables

For each merchant $g \in \{1, \ldots, G\}$, the model consumes the following variables over T weekly time steps across C marketing channels: the aggregated media spend $x_{t,c} \in \mathbb{R}_{\geq 0}$ of channel c in week t, a primary target $y_t \in \mathbb{R}$ (Net Gross Profit), an auxiliary target $y_t^{(int)} \in \mathbb{R}_{\geq 0}$ (total impressions), and a control feature vector $z_t \in \mathbb{R}^{19}$ comprising lagged targets and calendar indicators. The remainder of this section details each variable.

The primary target variable predicted by the model is defined as

$$y_t = \text{Net Gross Profit}_t \,, \tag{4.2}$$

which represents the net economic value generated during week t. Net Gross Profit is a proprietary metric used by the host organization, defined as gross revenue net of product returns and fulfillment costs. It provides a measure of operational profitability directly attributable to sales activity, excluding fixed overhead and marketing spend, a property that makes it a suitable target for marketing effectiveness modeling, as the cost of the marketing interventions being measured is not subtracted from the outcome variable.

In addition to the primary target, the model predicts an intermediate marketing indicator used to stabilize training:

$$y_t^{(int)} = \text{Impressions}_t \,. \tag{4.3}$$

Impression counts provide a lower-variance indicator of campaign activity than revenue, offering a denser supervision signal that helps the temporal attention module learn meaningful representations in the low-data regime. This multitask strategy follows the approach adopted in the NNN framework Mulc et al. (2025), which uses an analogous auxiliary target to stabilize temporal attention learning, and is consistent with the broader multitask learning literature Caruana (1997). The role of this auxiliary objective in the training loss is described in Section 4.2.

4.1.1.2 Temporal Aggregation

The raw dataset is originally available at daily resolution but is aggregated at the weekly level before model training. For a representative merchant, this produces approximately 200 weekly observations from over 1,400 daily records.

For each marketing channel c and week t, the weekly spend is computed as

$$x_{t,c} = \sum_{d \in week(t)} spend_{d,c} \,. \tag{4.4}$$

Weekly aggregation is widely used in MMM because it reduces intra-week noise while preserving the temporal dynamics relevant to marketing response modeling. This trade-off is particularly important in the present setting: with limited total observations, reducing variance at the cost of temporal resolution improves the stability of the learned attention profiles and the reliability of channel attribution estimates.

4.1.2 Hierarchical Channel Embedding

One of the key challenges in modern MMM settings is the limited number of temporal observations relative to the number of marketing channels. To improve statistical efficiency under these constraints, the model introduces a hierarchical representation of marketing channels.

Each channel is characterized by three structural levels:

- **Channel group** (e.g., social, search, direct)
- **Channel** (e.g., Facebook, Google, TikTok)
- **Funnel stage** (e.g., awareness, consideration, conversion)

For a representative merchant with 7 channels, 5 funnel positions, and 4 groups, these three levels describe 16 unique channel combinations using only $4+7+5 = 16$ independent embedding vectors, rather than one free vector per channel. The hierarchy is backward-compatible: configurations with only two levels (platform and funnel) remain fully supported, and the group level adds expressiveness only when cross-platform similarities are present in the data.

This factored representation, illustrated in Figure 4.3, allows the model to share statistical strength across channels with similar structural properties, acting as a regularizing constraint in low-data regimes.

For each channel c and time step t, the embedding vector is defined as

$$e(c,t) = E_{channel}(c) + E_{funnel}(c) + E_{group}(c) + f(spend_{c,t}) \,, \tag{4.5}$$

where $E_{channel}(c) \in \mathbb{R}^D$ is a learnable embedding associated with the marketing channel, $E_{funnel}(c) \in \mathbb{R}^D$ encodes the position of the channel within the marketing funnel, $E_{group}(c) \in \mathbb{R}^D$ represents the broader channel

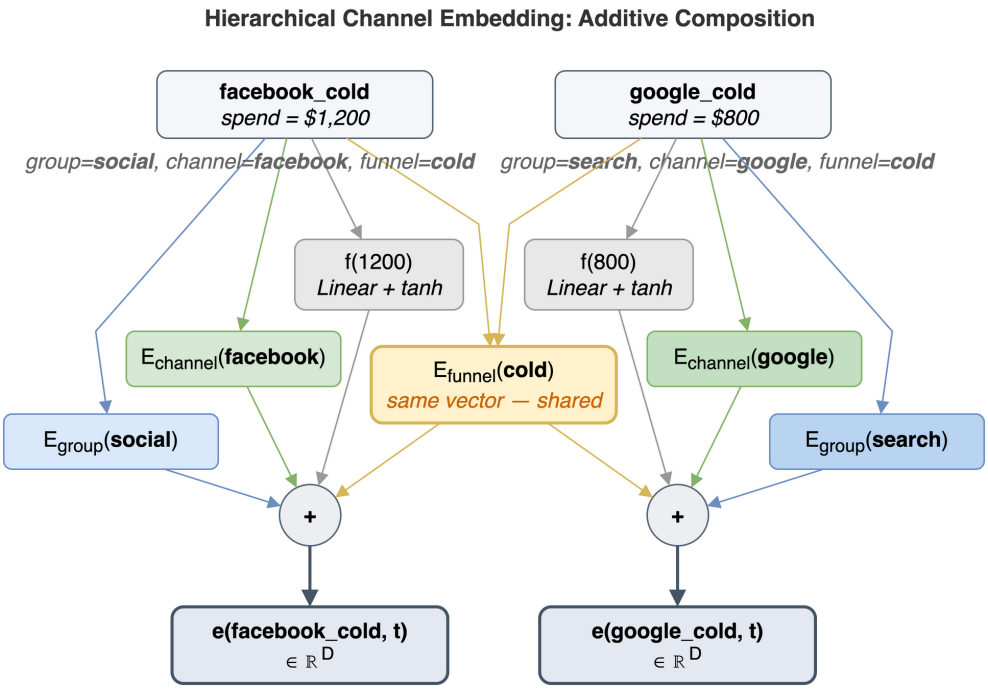


Figure 4.3: Hierarchical channel representation. Each marketing input (e.g., `facebook_cold`) is decomposed into three structural levels, each mapped to a learnable embedding vector. The funnel embedding is shared across channels with the same funnel position. The final channel embedding is the additive composition of these components together with a spend projection.

category, and $f(\cdot)$ is a projection of the observed spend value into the embedding space.

The spend projection is implemented using a linear transformation followed by the hyperbolic tangent activation:

$$f(spend_{c,t}) = \tanh(W_{spend} \cdot spend_{c,t} + b_{spend}), \tag{4.6}$$

where $W_{spend} \in \mathbb{R}^D$ is a learnable weight vector, $b_{spend} \in \mathbb{R}^D$ is a bias term, and $\tanh(x) = (e^x - e^{-x})/(e^x + e^{-x})$ maps each element to the interval $[-1, 1]$. This bounding prevents high-spend channels from dominating the embedding space and ensures that the magnitude of the spend component remains commensurate with the learned categorical embeddings.

The embedding is composed additively. This is a deliberate choice: while the additive formulation cannot capture non-linear interactions between hierarchy levels, it enforces a clean decomposition into group, channel, and funnel components, providing a regularization effect that is preferable to a more expressive but data-hungry alternative when observations are scarce.

To quantify the parametric efficiency of this design: with an embedding dimension $D = 32$ and a representative hierarchy of 7 channels, 5 funnel stages, and 4 groups, the embedding module uses $(7 + 5 + 4) \times 32 = 512$ parameters for the categorical components plus $32 + 32 = 64$ parameters for the spend projection (a linear layer with bias), totaling approximately 576 parameters. A flat embedding for 14 channels would require $14 \times 32 = 448$ parameters, comparable in count, but without the inductive bias that forces compositional structure and enables generalization across channels sharing the same channel identity, funnel, or group.

4.1.3 Multi-Scale Temporal Attention

Modeling the temporal dynamics of marketing effects is a central challenge in MMM. Different marketing channels exhibit heterogeneous response profiles: lower-funnel activities such as branded search may lead to immediate conversions, while upper-funnel campaigns such as awareness advertising may influence consumer behavior over longer time horizons.

Traditional MMM approaches address this problem through parametric adstock transformations or distributed lag models (Broadbent, 1979; Jin et al., 2017). However, these approaches typically require specifying a fixed functional form for the temporal decay process.

To allow greater flexibility, the proposed model employs a neural temporal attention mechanism that learns the temporal influence structure directly from

the data. Building upon the attention-based architecture proposed in Mulc et al. (2025), the model introduces a *multi-scale temporal attention* mechanism designed to capture marketing effects occurring at different time horizons simultaneously. This extension is consistent with the recommendation of the original NNN authors, who note in Section 13.1 of their paper: "Implementing multi-head attention, rather than the current single-head approach, might allow different heads to specialize in distinct temporal patterns" Mulc et al. (2025). The multi-scale design proposed here realizes this suggestion by assigning each head a different temporal window.

4.1.3.1 Multi-scale architecture

Let $X \in \mathbb{R}^{G\times T\times C\times D}$ denote the embedded input tensor, where G is the number of geographical units or merchants, T the number of time periods, C the number of marketing channels, and D the embedding dimension.

Instead of using a single temporal attention mechanism, the model processes the input through multiple temporal attention heads operating at different temporal scales.

Each scale corresponds to a different temporal window ω_i, capturing marketing responses occurring at different time horizons:

$$\omega_i \in \{1, 2, 4, 8\} \text{ weeks}\,. \tag{4.7}$$

For each scale i, a temporal attention block produces an output representation $H_i \in \mathbb{R}^{G\times T\times C\times D}$:

$$H_i = \text{TemporalAttention}_{\omega_i}(X)\,, \tag{4.8}$$

where X is the embedded input tensor defined above. The final representation is obtained as a weighted combination of the scale-specific outputs:

$$H = \sum_{i=1}^{N} \alpha_i H_i\,, \tag{4.9}$$

where N is the number of scales and α_i are non-negative weights summing to one, computed from learnable logits θ_i via a softmax transformation:

$$\alpha_i = \frac{\exp(\theta_i)}{\sum_{j=1}^{N} \exp(\theta_j)}\,. \tag{4.10}$$

This mechanism allows the model to automatically learn which temporal scales are most relevant for the observed marketing dynamics.

4.1.3.2 Per-channel temporal attention

Within each temporal scale, the model assigns an independent temporal attention function to each channel. The original NNN architecture Mulc et al. (2025) uses a single shared MLP for all channels, which works well with two channels but produces nearly identical decay profiles when the number of channels increases to 9–14, as gradient updates from all channels compete for the same parameters. To remove this bottleneck, each channel c receives its own small MLP ($1 \to 16 \to 1$, 49 parameters) that maps the normalized time lag to an attention score:

$$s_c(t,t') = \mathrm{MLP}_c\left(\frac{t'-t}{\omega}\right), \qquad t' < t\,, \tag{4.11}$$

where c indexes the channel and the constraint $t' < t$ enforces causality, ensuring that only past time steps contribute to the representation at time t. Scores are normalized via softmax with temperature τ to produce attention weights, and the temporal representation is computed as a weighted aggregation of past embeddings:

$$W_{time}(t,c,t') = \frac{\exp(s_c(t,t')/\tau)}{\sum_{t''}\exp(s_c(t,t'')/\tau)}, \qquad H_{t,c} = \sum_{t'} W_{time}(t,c,t')\; X_{t',c}\,. \tag{4.12}$$

Figures 4.4–4.5 provide a visual reference for the temporal attention mechanism described above, using a representative merchant. Figure 4.4 shows the learned scale-weight distribution: the model allocates most weight to the 1-week scale, with progressively lower weights for longer horizons. Figure 4.5 shows the effective cumulative temporal response per channel, obtained as the scale-weighted combination across all heads, the neural equivalent of the parametric adstock functions used in traditional MMM. The diversity of shapes confirms that the per-channel multi-scale design learns qualitatively distinct response dynamics.

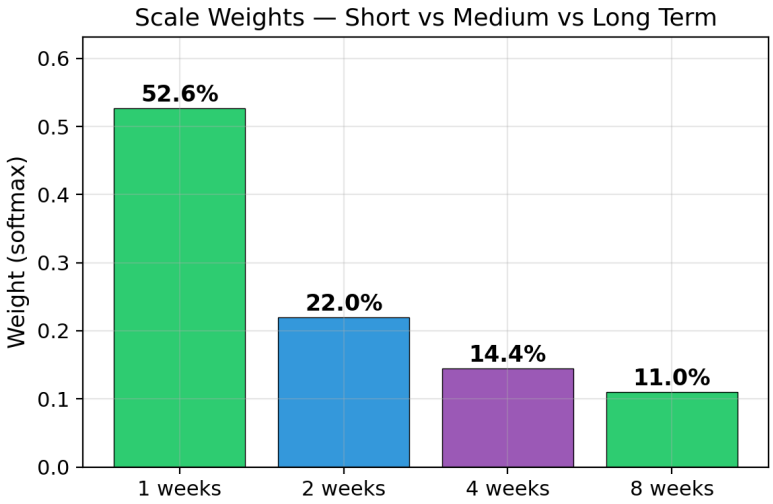


Figure 4.4: Learned scale-weight distribution for a representative merchant. The 1-week scale receives 52.6% of the total weight, with progressively lower weights for longer horizons.

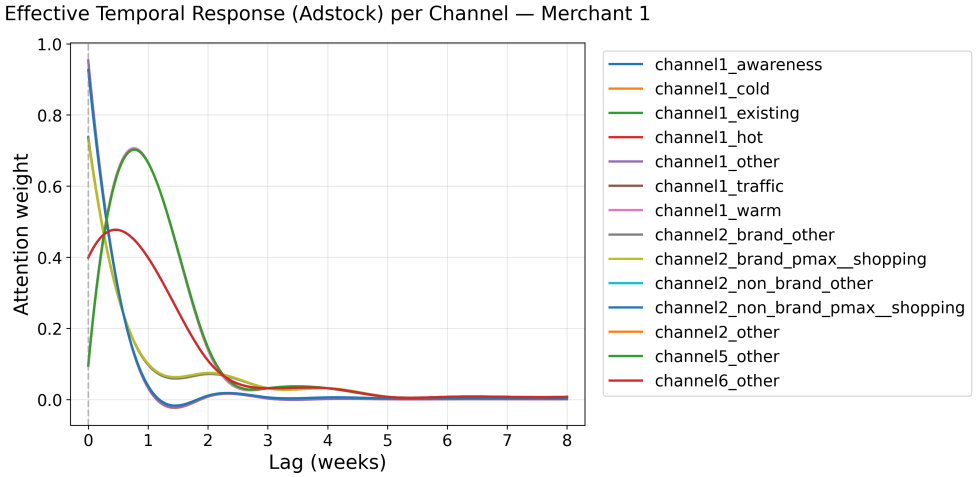


Figure 4.5: Effective cumulative temporal response (adstock) per channel for a representative merchant, obtained as the scale-weighted combination of per-channel attention profiles. Different channels exhibit qualitatively distinct dynamics, from sharp immediate decay to sustained delayed effects.

4.1.4 Cross-Channel Interaction Modeling

In addition to modeling the temporal dynamics of individual marketing channels, it is also necessary to account for interactions between channels. In real-world marketing systems, channels rarely operate in isolation. Instead, they often influence each other through complex demand propagation mechanisms along the marketing funnel.

For example, upper-funnel campaigns may increase brand awareness, which later manifests as increased branded search queries or direct traffic. Such effects introduce delayed dependencies between channels that cannot be captured by purely channel-independent temporal response functions.

Traditional MMM approaches typically assume additive channel effects or rely on predefined interaction terms Naik and Raman (2003). However, these approaches require manually specifying the interaction structure and cannot easily capture dynamic lagged dependencies.

To address this limitation, the proposed model introduces a learnable cross-channel interaction mechanism that models the influence of one channel on another across time.

4.1.4.1 Static channel interaction matrix

Let C denote the number of marketing channels. The model first introduces a learnable interaction matrix

$$S \in \mathbb{R}^{C \times C}, \tag{4.13}$$

where each element $S_{c,v}$ represents the influence of channel v on channel c.

The diagonal elements $S_{c,c}$ correspond to self-influence, while off-diagonal elements capture cross-channel interactions.

The interaction weights are normalized through a softmax operation:

$$W_{channel}(c, v) = \frac{\exp(S_{c,v})}{\sum_k \exp(S_{c,k})}. \tag{4.14}$$

This matrix allows the model to redistribute attention across channels when computing the temporal representation.

4.1.4.2 Lag-dependent interaction effects

While the static interaction matrix captures instantaneous relationships between channels, many marketing interactions exhibit delayed effects.

To model such dynamics, the interaction scores are extended with a lag-dependent component based on a parametric impulse response function.

For each pair of channels (c, v) and time lag τ, the interaction score is defined as

$$score(c, v, \tau) = S_{c,v} + \beta_{c,v} \cdot K(\tau)\,, \tag{4.15}$$

where $S_{c,v}$ is the base interaction strength, $\beta_{c,v}$ controls the amplitude of the lagged interaction, and $K(\tau)$ is a temporal kernel describing how influence evolves over time.

In the proposed model, the kernel $K(\tau)$ is parameterized using a Weibull distribution.

4.1.4.3 Weibull impulse response kernel

The Weibull distribution (Weibull, 1951) provides a flexible functional form capable of representing both monotonic decay and peaked response curves.

The kernel is defined as

$$K(\tau; k, \lambda) = \frac{k}{\lambda}\left(\frac{\tau}{\lambda}\right)^{k-1} \exp\left(-\left(\frac{\tau}{\lambda}\right)^{k}\right), \tag{4.16}$$

where $k > 0$ is the shape parameter and $\lambda > 0$ is the scale parameter. Different values of k produce qualitatively distinct response dynamics: $k < 1$ yields monotonically decreasing decay, $k \approx 2$ produces a delayed peak response, and $k > 2$ generates sharply peaked patterns.

This flexibility allows the model to capture a wide range of cross-channel demand propagation patterns observed in marketing systems.

The Weibull kernel was selected over several alternatives. Gaussian basis functions offer similar expressiveness but require choosing the number and spacing of basis centers, introducing discrete hyperparameters that are difficult to validate in low-data settings. Vector autoregressive (VAR) models Sims (1980) would require $O(C^2 \cdot L)$ parameters for L lags, which is prohibitive with 9–14 channels and limited observations. Hawkes-process formulations Mei and Eisner (2017) assume event-driven dynamics that do not naturally fit the aggregated weekly spending data used in MMM. The Weibull parametrization requires only two parameters per channel pair (k and λ), is differentiable end-to-end, and admits a direct marketing interpretation: λ controls when the cross-channel effect peaks, and k controls whether the response is immediate or delayed. This asymmetry in model complexity, high capacity for within-channel temporal dynamics (per-channel MLPs)

paired with a parsimonious parametric form for cross-channel effects, reflects the prior that self-channel carryover patterns are richer and more reliably estimable than inter-channel spillovers.

4.1.4.4 Channel interaction weights

The final interaction weights are obtained by normalizing the lag-dependent scores across channels:

$$W_{channel}(c, v, \tau) = \frac{\exp(score(c, v, \tau)/\tau_{ch})}{\sum_{v'} \exp(score(c, v', \tau)/\tau_{ch})}, \tag{4.17}$$

where τ_{ch} is a temperature parameter controlling the smoothness of the channel attention distribution.

4.1.4.5 Inductive bias and prior initialization

Learning cross-channel relationships purely from data can be unstable in MMM settings due to the limited number of observations and the relatively large number of parameters.

To mitigate this issue, the model incorporates the following initialization strategy.

The diagonal elements of the interaction matrix S are initialized to 3.0, encouraging the model to initially focus on self-channel carryover before learning cross-channel effects. Off-diagonal entries are initialized with small uniform values, so that all pairwise interactions start from a low but non-zero baseline. The amplitude parameter β is initialized to zero for all pairs, ensuring that no cross-channel temporal modulation is assumed before training. The Weibull shape k and scale λ parameters are initialized with a mild recency bias (values favoring short-delay peaks), reflecting the prior that most cross-channel effects attenuate within a few weeks.

These mechanisms help stabilize training and improve interpretability of the learned interaction patterns.

Figures 4.6–4.7 provide a visual reference for the cross-channel interaction mechanism using a representative merchant. Figure 4.6 shows the lag-dependent influence share for the channel pairs with the strongest temporal variation: each curve represents how the attention share from a source channel to a target channel evolves over time, with the dashed line indicating the uniform baseline (no interaction). Figure 4.7 shows two individual pairs in detail, decomposing the learned interaction into a static baseline share and the temporal effect added by the Weibull kernel. The two examples illustrate

different delay dynamics: one within-platform pair peaks at +1 week, while a second platform's equivalent pair peaks at +2 weeks with a broader profile.

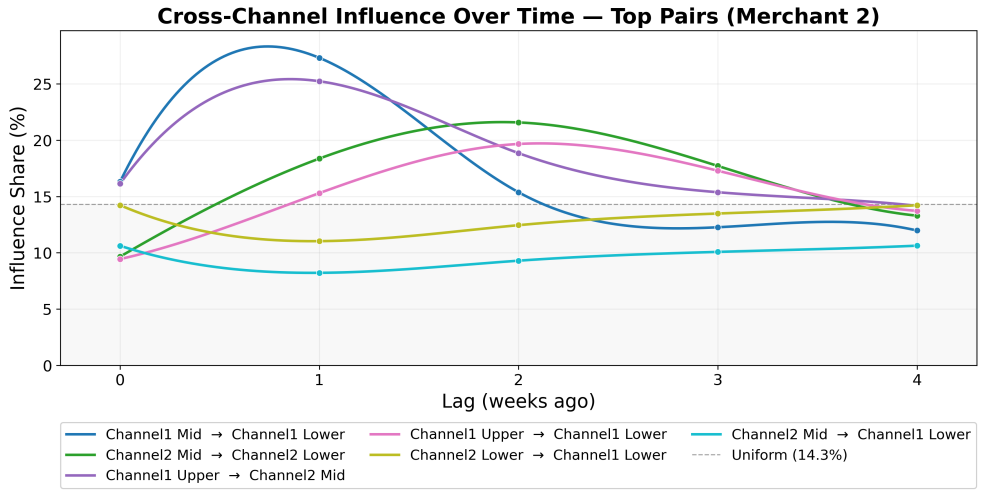


Figure 4.6: Cross-channel influence over time for the top channel pairs by temporal variation (representative merchant). Each curve shows the attention share from a source to a target channel as a function of lag. The dashed line indicates the uniform share (14.3% with 7 channels), above which the Weibull kernel adds a positive temporal interaction effect.

4.1.5 Factored attention combination

Following the factored self-attention design introduced in Mulc et al. (2025), the temporal and cross-channel attention mechanisms are not applied sequentially but combined through an element-wise product. For a given time step t and target channel c, the combined attention weight assigned to channel v at lag τ is

$$W(t, c, v, \tau) = W_{time}(t, c, \tau) \cdot W_{channel}(c, v, \tau)\,. \tag{4.18}$$

This factorization has a precise semantic interpretation: W_{time} determines *when* past information is relevant (temporal selection), while $W_{channel}$ determines *from which channels* information should flow (structural selection). The multiplicative combination ensures that a past observation receives high attention only if it is relevant both temporally and structurally, without requiring the model to learn the full joint distribution over time steps and channels, which would involve $O(T \times C)$ parameters per query and would be impractical in low-data settings.

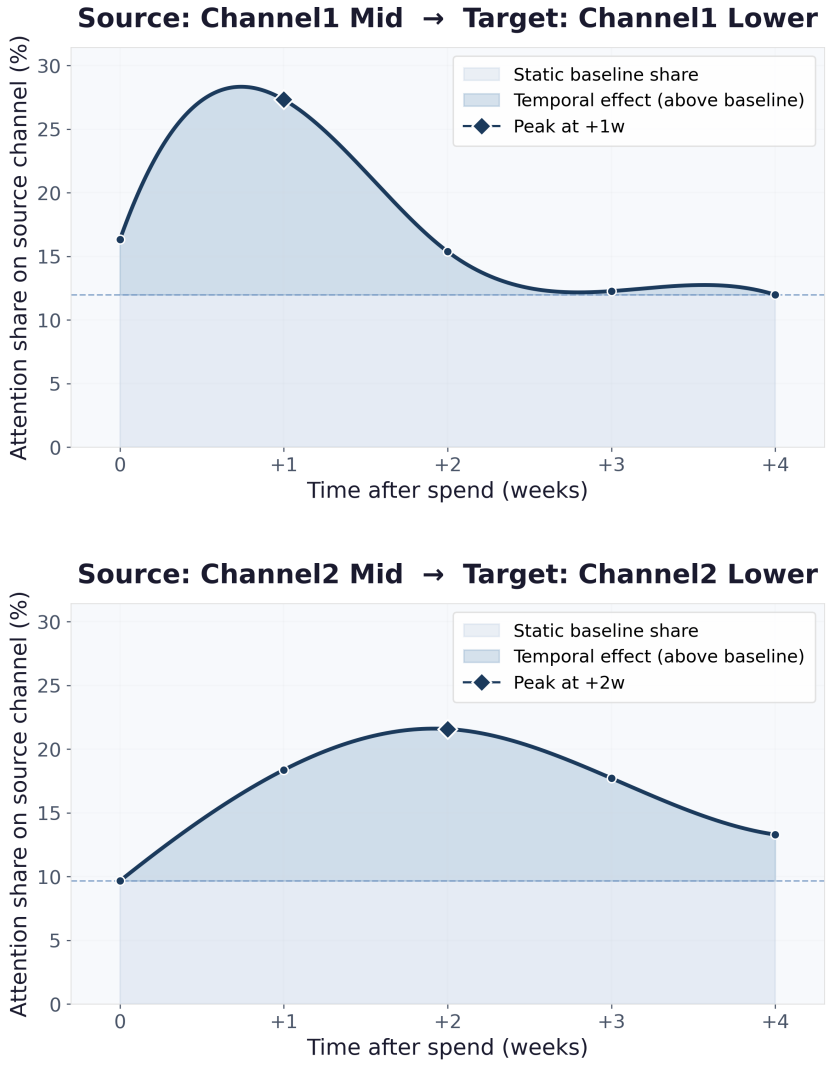


Figure 4.7: Detailed view of two individual cross-channel interactions. **Top:** within-platform mid-funnel → lower-funnel, peaking at +1 week, where mid-funnel activity on the same platform produces a rapid spillover into lower-funnel conversions. **Bottom:** a second platform's mid-funnel → lower-funnel, peaking at +2 weeks with a slower pipeline effect with a broader temporal profile. The shaded area above the dashed baseline shows the temporal contribution of the Weibull kernel.

The enriched representation for channel c at time t is then computed as

$$H_{t,c} = \sum_{v} \sum_{\tau} W(t, c, v, \tau)\ X_{t-\tau,v}\,, \tag{4.19}$$

aggregating information across both the temporal and channel dimensions according to the learned factored weights.

4.1.6 Prediction Heads and Baseline Decomposition

After the temporal and cross-channel interaction mechanisms have produced a latent representation of the marketing signals, the model must transform these representations into interpretable predictions.

In MMM, interpretability is critical: the model must not only predict the target variable but also provide a decomposition of the predicted outcome into channel-specific contributions and an organic baseline.

To achieve this objective, the proposed architecture separates the prediction stage into a *per-channel media prediction head*, which estimates the contribution of each marketing channel, and a *baseline module*, which captures non-marketing demand dynamics. The final predicted outcome is obtained as the sum of these two components.

4.1.6.1 Per-channel media contribution

Let $H_{t,c} \in \mathbb{R}^D$ denote the latent representation of channel c at time t produced by the attention module.

Instead of aggregating channels before prediction, the model applies a dedicated prediction head to each channel representation.

The contribution of channel c is defined as

$$effect_{t,c} = g_c(H_{t,c})\,, \tag{4.20}$$

where $g_c(\cdot)$ is a channel-specific linear projection followed by a non-linear activation.

In practice, this transformation is implemented as

$$effect_{t,c} = \text{Softplus}(w_c^\top H_{t,c} + b_c)\,, \tag{4.21}$$

where w_c and b_c are learnable parameters.

The Softplus activation function ensures that predicted channel contributions remain non-negative:

$$\text{Softplus}(x) = \log(1 + e^x)\,. \tag{4.22}$$

This constraint reflects the assumption that marketing investments should not generate negative incremental effects on the target variable. Similar non-negativity assumptions are commonly adopted in MMM models to improve interpretability Jin et al. (2017).

4.1.6.2 Baseline component

In addition to media-driven effects, a large portion of sales or revenue is typically driven by organic demand dynamics unrelated to marketing campaigns, including accumulated brand awareness, seasonal purchasing behavior, macroeconomic factors, and repeat customer demand. To capture these dynamics, the model introduces a separate baseline component.

The baseline module receives two types of inputs: lagged values of the target variable, which encode recent demand momentum, and temporal features consisting of one-hot encoded month and day-of-week indicators (19 dimensions in total), which capture seasonal and weekly patterns.

Let Z_t denote the vector containing these baseline features. The baseline prediction is computed through a small neural network:

$$baseline_t = (\text{ELU} + 1)(f_{base}(Z_t))\,, \tag{4.23}$$

where f_{base} is implemented as a multi-layer perceptron and the ELU + 1 output activation ensures that the baseline remains non-negative, consistent with the non-negativity constraint applied to media contributions. The ELU + 1 function is preferred over ReLU because it has non-zero gradients for negative inputs, avoiding dead neurons during early training when the baseline module has not yet learned to produce positive outputs.

Importantly, the baseline module does not receive any direct information about media spend. This separation helps ensure that media contributions are not absorbed by the baseline component, preserving the interpretability of the channel-level decomposition.

4.1.6.3 Final prediction

The final model prediction combines the baseline component and the aggregated media effects:

$$\hat{y}_t = \alpha_{base} \cdot baseline_t + media_t\,, \tag{4.24}$$

where α_{base} is a learnable scalar initialized to one. Because the baseline MLP and the media prediction heads are trained jointly from random initialization, their output magnitudes can differ significantly in the early training phases. Rather than relying on the baseline MLP to internally learn the correct scale, α_{base} provides a single degree of freedom that lets the optimizer quickly balance the two branches. At convergence, α_{base} typically settles close to one, confirming that the two branches have learned compatible scales.

This additive decomposition mirrors the standard MMM formulation $y_t = baseline_t + \sum_c media_{t,c}$, providing a direct interpretation of the output as the sum of organic demand and marketing-driven components.

4.2 Training Objective

The model is trained with a composite objective that combines supervised multitask prediction, parameter regularization, and structural regularization terms that address specific failure modes of the multi-scale temporal attention mechanism.

4.2.1 Primary prediction loss

Let $\hat{y}_t$ denote the predicted value of Net Gross Profit at week t, and let y_t denote the corresponding observed target. The primary prediction loss is defined as a mean squared error:

$$\mathcal{L}_{sales} = \frac{1}{T} \sum_{t=1}^{T} (y_t - \hat{y}_t)^2 \ . \tag{4.25}$$

Mean squared error is adopted because the task is formulated as a continuous regression problem and because it strongly penalizes large deviations between predicted and observed business outcomes.

4.2.2 Auxiliary multitask loss

In addition to the primary target, the model predicts an intermediate signal related to advertising exposure, denoted by $y_t^{(int)}$. Let $\hat{y}_t^{(int)}$ be the corresponding prediction produced by the intermediate head. The auxiliary loss is defined as

$$\mathcal{L}_{int} = \frac{1}{T} \sum_{t=1}^{T} \left(y_t^{(int)} - \hat{y}_t^{(int)} \right)^2 \ . \tag{4.26}$$

As discussed at the beginning of this section, the auxiliary objective provides a denser supervision signal that regularizes the temporal attention mechanism in the low-sample regime. Both the choice of an intermediate marketing indicator as auxiliary target and the multitask formulation follow the NNN framework Mulc et al. (2025).

4.2.3 Joint objective

The overall training objective combines the two losses through a weighted sum:

$$\mathcal{L}_{task} = \alpha_{sales}\mathcal{L}_{sales} + \alpha_{int}\mathcal{L}_{int}\,, \tag{4.27}$$

where α_{sales} and α_{int} are non-negative coefficients controlling the relative importance of the two tasks.

Because the primary goal of the model is to predict economic performance, the main task receives the dominant weight, while the intermediate objective acts as a secondary regularizing signal. This design ensures that the model remains focused on the business outcome of interest while still benefiting from the additional structure provided by the auxiliary target.

4.2.4 Parameter regularization

Given the limited size of typical MMM datasets, controlling model complexity is essential to reduce overfitting. For this reason, parameter shrinkage is introduced through an L_1 regularization term:

$$\mathcal{L}_{reg} = \lambda \sum_j |w_j|\,, \tag{4.28}$$

where w_j denotes a trainable model parameter and λ controls the strength of the penalty.

L_1 regularization encourages sparse and stable solutions across the model's high-capacity components, including channel embeddings, temporal attention functions, and cross-channel interaction parameters. This regularization strategy is adopted from the NNN framework Mulc et al. (2025).

4.2.5 Temporal diversity loss

Without explicit regularization, per-channel temporal MLPs may converge to nearly identical profiles for channels with weak spend signals, whose temporal structure contributes little to the primary loss and is therefore flattened by

the optimizer. The diversity loss penalizes high cosine similarity between the effective temporal profiles of different channels:

$$\mathcal{L}_{div} = \frac{1}{C(C-1)} \sum_{c \neq c'} \cos_\text{sim}(\tilde{p}_c, \tilde{p}_{c'}) \,, \tag{4.29}$$

where $\tilde{p}_c = \sum_i \alpha_i \cdot p_{c,i}$ is the scale-weighted temporal profile of channel c, obtained by combining the per-scale attention weights $p_{c,i}$ with the learned scale logits α_i. This term encourages the model to learn distinct temporal behaviors when the data support differentiation.

4.2.6 Final loss function

The complete training objective combines all components:

$$\mathcal{L} = \underbrace{\alpha_{sales}\mathcal{L}_{sales} + \alpha_{int}\mathcal{L}_{int}}_{\text{prediction}} + \underbrace{\lambda\mathcal{L}_{reg}}_{\text{sparsity}} + \underbrace{\lambda_{div}\mathcal{L}_{div}}_{\text{temporal structure}} \,. \tag{4.30}$$

Table 4.1 summarizes the loss coefficients and their typical values.

Table 4.1: Loss function coefficients and typical values.

Coefficient	Component	Typical value	Effect
α_{sales}	Primary MSE	0.8	Revenue prediction weight
α_{int}	Auxiliary MSE	0.2	Impression prediction weight
λ	Global L_1	0.001–0.1	Weight sparsity
λ_{div}	Diversity loss	0.01–0.1	Profile differentiation

4.3 Training Procedure

The model is trained using a supervised procedure that preserves the temporal structure of the data. The dataset is split chronologically into training, validation, and test sets with a typical allocation of 70%/15%/15%, enforcing the strict ordering constraint $t_{train} < t_{val} < t_{test}$ to prevent information leakage from future observations. Figure 4.8 illustrates this split for a representative merchant after normalization.

Parameters are optimized with Adam Kingma and Ba (2015), using an initial learning rate selected through hyperparameter search (typically in

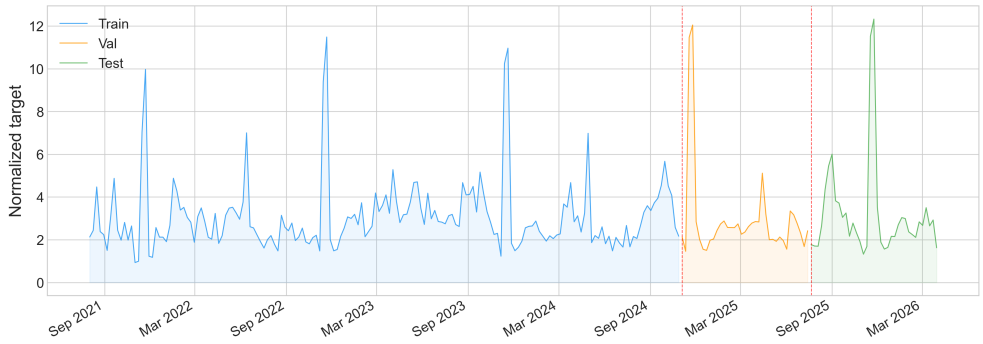


Figure 4.8: Chronological train/validation/test split (70/15/15) for a representative merchant. The target variable is shown after global linear normalization. Red dashed lines mark the split boundaries.

$[0.001, 0.05]$). A ReduceLROnPlateau scheduler monitors validation loss and reduces the learning rate by a factor of 0.5 if no improvement is observed for 100 consecutive epochs. Training terminates via early stopping (Prechelt, 2012) with a patience window of 200–300 epochs; the model weights from the best validation epoch are restored. The maximum training budget is set to 2,000–5,000 epochs, but effective training typically completes in 500–1,500 epochs.

4.3.1 Training rationale

The training procedure balances three requirements central to the MMM setting: (i) temporal integrity through chronological splitting, preventing leakage from future observations; (ii) optimization flexibility through Adam with adaptive learning-rate scheduling, accommodating a multi-component architecture where embeddings, attention, and prediction heads converge at different rates; and (iii) overfitting control through early stopping with weight restoration, complementing the explicit regularization terms in the loss function.

4.4 Hyperparameter Selection

The model exposes approximately 30 configurable parameters, which fall into three categories. A first group is determined by the merchant's specific context (channel hierarchy, number of geos, funnel definitions) and is set once during onboarding. A second group defines the model's architectural identity

and is fixed across all experiments (Table 4.2). The remaining parameters control optimization dynamics and model capacity; after an initial manual calibration informed by training diagnostics, they are refined through Bayesian optimization (Snoek et al., 2012), implemented using the Optuna framework (Akiba et al., 2019). Searching all of them jointly would produce an intractably large space given the limited data, so the optimization is split into two focused phases of 3–4 parameters each.

Table 4.2: Fixed architectural parameters (not searched).

Parameter	**Value**	**Rationale**
Temporal attention mode	multi_scale	Core thesis contribution
Hierarchy embedding	enabled	Channel representation
Per-channel sales heads	enabled	Signal Reconstruction
Normalization	global scale	Preserves magnitude ratios
Temporal initialization	funnel-aware decay	Domain knowledge prior
Channel mixing	enabled	Cross-channel modeling

4.4.1 Phase 1: Stabilization

The first phase identifies convergence parameters that produce a model with positive R^2 on the validation set. Three parameters are searched via Bayesian optimization over 30–50 trials: learning rate $\in [0.001, 0.05]$ (log scale), L_1 regularization strength $\lambda \in [0.001, 0.1]$ (log scale), and baseline weight $\alpha_{base} \in [0.3, 5.0]$.

4.4.2 Phase 2: Capacity

The second phase determines the appropriate model complexity. Four parameters are searched while holding the Phase 1 results fixed: embedding dimension $D \in \{8, 16, 32\}$, hidden dimension $h \in \{16, 32, 64\}$, maximum lookback window $\omega \in \{7, 14, 21, 28\}$ weeks, and head dropout (Srivastava et al., 2014) $\in [0.1, 0.5]$. This sequential strategy exploits the observation that convergence parameters interact weakly with capacity parameters.

4.4.3 Phase 3: Baseline

The third phase tunes the baseline component, which models organic (non-media) demand. Its optimal configuration depends on what the media-attribution part of the network has already learned, so it is searched last. Three

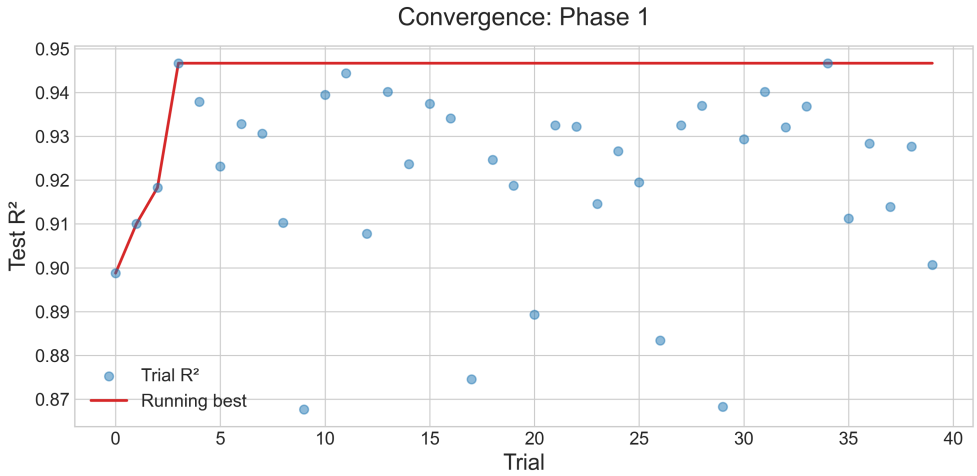


Figure 4.9: Phase 1 (Stabilization): Bayesian optimization progress for a representative merchant. The plot shows validation R^2 across trials as the optimizer explores learning rate, L_1 strength, and baseline weight. Detailed parameter distributions are reported in Appendix A.

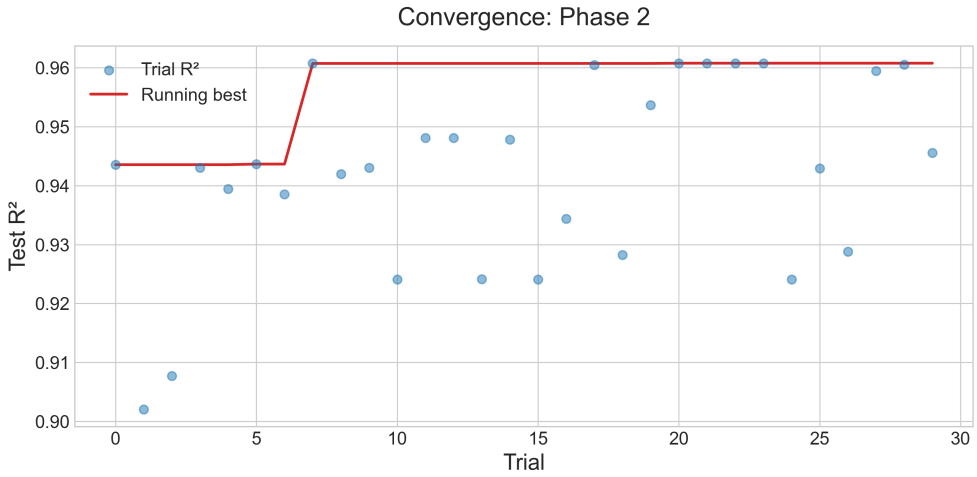


Figure 4.10: Phase 2 (Capacity): Bayesian optimization progress for a representative merchant. The plot shows validation R^2 across trials as the optimizer explores embedding dimension, hidden dimension, lookback window, and dropout. Detailed parameter distributions are reported in Appendix A.

parameters are optimized: baseline weight $\alpha_{base} \in [0.3, 3.0]$ (re-searched in the context of the tuned model), lag option (an integer index selecting which set of autoregressive lags to use as baseline features), and initial bias $\in [0.5, 5.0]$ (the starting value of the baseline prediction head).

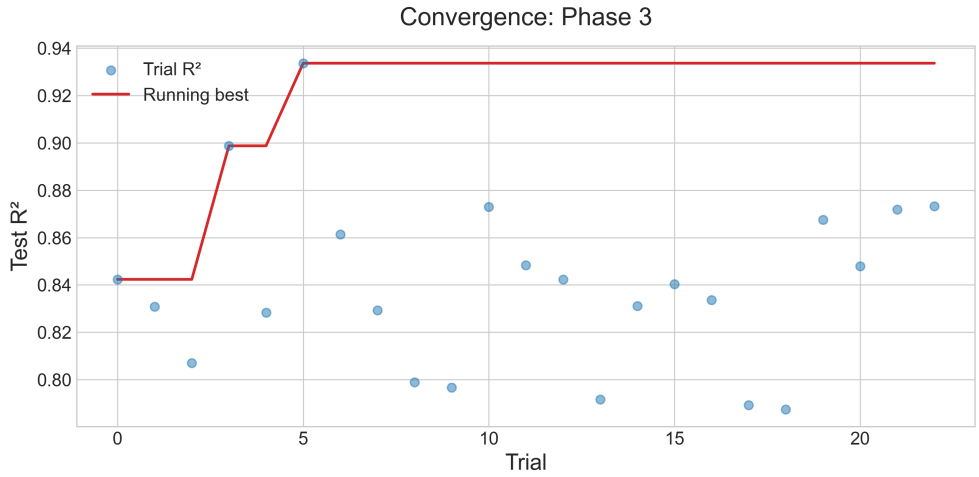


Figure 4.11: Phase 3 (Baseline): Bayesian optimization progress for a representative merchant. The plot shows validation R^2 across trials as the optimizer explores baseline weight, lag configuration, and initial bias. Detailed parameter distributions are reported in Appendix A.

A small number of parameters with strong domain interactions, namely temporal scales (e.g., $[1, 2, 4, 8]$ weeks, matched to the merchant's marketing cadence), channel attention prior (diagonal initialization strength), and multitask balance (α_{sales}, α_{int}), are set manually based on training diagnostics rather than searched automatically.

4.5 Attribution Method

Channel attribution is computed through a counterfactual leave-one-out procedure. For each media channel c, the model is re-evaluated with that channel's spend zeroed out across all time steps, and the attribution is defined as the difference in predicted outcomes:

$$A_{t,c} = \hat{y}_t - \hat{y}_t|_{X_c=0}, \tag{4.31}$$

where $\hat{y}_t|_{X_c=0}$ denotes the prediction when channel c receives zero input. This requires $M + 1$ forward passes and captures both direct

channel contributions and indirect effects mediated through the cross-channel interaction kernel.

Negative attributions (removing a channel increases predicted sales) are clamped to zero, as they typically indicate model instability rather than a genuine negative marketing effect. Cumulative attribution over the observation period is $A_c = \sum_t A_{t,c}$ and can be used to derive aggregate indicators such as return on advertising spend (ROAS).

4.6 Implementation Summary

This chapter presented the implementation of the neural architecture proposed for the MMM task. Table 4.3 summarizes the main implementation characteristics for a representative merchant configuration.

Table 4.3: Implementation summary for a representative merchant configuration.

Characteristic	**Value**
Framework	PyTorch
Total trainable parameters	∼10,000–15,000
Embedding dimension (D)	8–32
Number of temporal scales	3–4
Per-channel MLP parameters	49 per channel
Weibull kernel parameters	3 per channel pair
Hierarchical embedding parameters	∼576 (representative)
Typical training epochs	500–1,500 (early stopping)
Training time	∼2–5 minutes (CPU)
Hyperparameter search	∼70–100 Bayesian trials (three phases)

Relative to the original NNN architecture, the implementation introduces four principal extensions motivated by the transition from a two-channel synthetic setting to a real-world environment with 9–14 channels and limited observations:

1. A *hierarchical channel embedding* that factorizes channel identity into channel, funnel, and group components, providing an additive inductive bias with approximately 576 parameters instead of a flat embedding.

2. *Per-channel temporal MLPs* replacing the shared one-hot-conditioned MLP, resolving the representational bottleneck that causes profile averaging when the number of channels increases.

3. A *multi-scale factored attention* mechanism with four temporal windows, realizing the multi-head extension suggested by the NNN authors, combined with learned scale weights regularized through a KL divergence prior and a temporal diversity loss.

4. A *Weibull-parametrized cross-channel interaction kernel* that adds lag-dependent spillover modeling with only two parameters per channel pair, with diagonal self-attention prior and mild recency initialization.

The prediction stage decomposes the outcome into non-negative per-channel media contributions (Softplus activation) and a non-negative organic baseline (ELU+1 activation), preserving the additive decomposition required by standard MMM interpretation. The model is trained with a composite objective combining multitask prediction, L_1 parameter regularization, temporal diversity loss. Training preserves temporal ordering through chronological splitting and uses validation-based early stopping. Hyperparameters are selected through a three-phase Bayesian optimization strategy that decomposes the search space into convergence and capacity parameters.

The next chapter evaluates the empirical performance of this architecture and compares it with traditional MMM approaches, analyzing predictive performance and structural behavior, including temporal dynamics, cross-channel interactions, and attribution decomposition.

Chapter 5

Results and Analysis

This chapter presents the empirical results of the comparative evaluation between the proposed attention-based neural architecture and the production Bayesian Marketing Mix Model. The analysis is organized around the two evaluation dimensions introduced in Chapter 1: predictive comparability and richer structural analysis of temporal and cross-channel marketing dynamics.

Section 5.1 summarizes the experimental configuration. Sections 5.2 and 5.3–5.4 present the main results. The chapter concludes with the reliability and validity analyses required by the evaluation framework.

5.1 Experimental Setup

The evaluation is conducted on three e-commerce merchants operating in the Nordic and European markets. Table 5.1 summarizes the key characteristics of each dataset. All merchants share the same data pipeline and model architecture described in Chapter 4; merchant-specific parameters (channel hierarchy, number of geos) are fixed by context, while architectural and optimization hyperparameters are tuned independently per merchant through the Bayesian search described in Section 4.4.

The baseline model is a production-grade Bayesian Marketing Mix Model deployed at the host company, built on the PyMC-Marketing framework with a custom extension layer. It uses parametric adstock transformations (geometric or Weibull decay, configurable per merchant) with per-channel shape parameters estimated via fully Bayesian inference. The model includes a modified Michaelis-Menten saturation function, Fourier seasonality components, and an optional time-varying baseline via Hilbert Space Gaussian Processes. Channel effects are assumed additive and independent, hence no cross-channel

Table 5.1: Dataset characteristics for the three evaluation merchants.

	Merchant 1	**Merchant 2**	**Merchant 3**
Marketing channels	14	9	14
Weekly observations	∼240	∼140	∼220
Train / Val / Test	70 / 15 / 15	70 / 15 / 15	70 / 15 / 15
Primary target	Net Gross Profit	Net Gross Profit	Net Gross Profit
Auxiliary target	Impressions	Impressions	Impressions

interaction mechanism is included. This system has been iteratively refined over multiple deployment cycles and represents a mature industry-standard benchmark. Since the Bayesian MMM produces a full posterior predictive distribution, all reported predictions and metrics correspond to the posterior mean, consistent with the point estimate used in the production deployment. Both models are evaluated on the held-out test set (the most recent 15% of observations) using two complementary metrics: the coefficient of determination (R^2) and the Weighted Mean Absolute Percentage Error (WMAPE). R^2 captures overall variance explained, while WMAPE provides an interpretable measure of prediction accuracy relative to observed magnitudes.

5.2 Predictive Performance

5.2.1 Overall Accuracy

Table 5.2 reports the test-set accuracy for the neural model and the Bayesian MMM across the three merchants. The neural model achieves R^2 values between 0.84 and 0.93, with WMAPE between 15.2% and 20.5%.

Table 5.2: Test-set predictive accuracy: Neural model vs Bayesian MMM.

	Neural Model		**Bayesian MMM**	
	R^2	WMAPE	R^2	WMAPE
Merchant 1	0.933	15.2%	0.930	16.2%
Merchant 2	0.903	17.4%	0.911	6.4%
Merchant 3	0.841	20.5%	0.921	14.6%

The results vary across merchants. On Merchant 1, the two models are nearly equivalent ($R^2 = 0.930$ vs 0.933; WMAPE 16.2% vs 15.2%),

with the neural model slightly ahead on both metrics. On Merchant 2, both models achieve similar R^2 (0.911 vs 0.903), but the Bayesian MMM produces a substantially lower WMAPE (6.4% vs 17.4%). On Merchant 3, the Bayesian MMM achieves clearly higher accuracy ($R^2 = 0.921$ vs 0.841; WMAPE 14.6% vs 20.5%). Overall, the Bayesian MMM tends to be more accurate, which is expected: it is a production system that has been iteratively refined over multiple deployment cycles, while the neural model is a research prototype evaluated without comparable tuning history. The accuracy gap, while meaningful, does not invalidate the structural analysis presented in the following sections; the neural model captures sufficient variance to produce interpretable temporal and cross-channel patterns.

5.2.2 Prediction Over Time

Figures 5.1–5.4 show the prediction diagnostics for the best-performing merchant (Merchant 1, where the neural model matches the Bayesian MMM) and the worst-performing merchant (Merchant 3, where the gap is largest). Each figure includes the predicted-vs-actual time series, a scatter plot with R^2, the error distribution, and a weekly bar comparison.

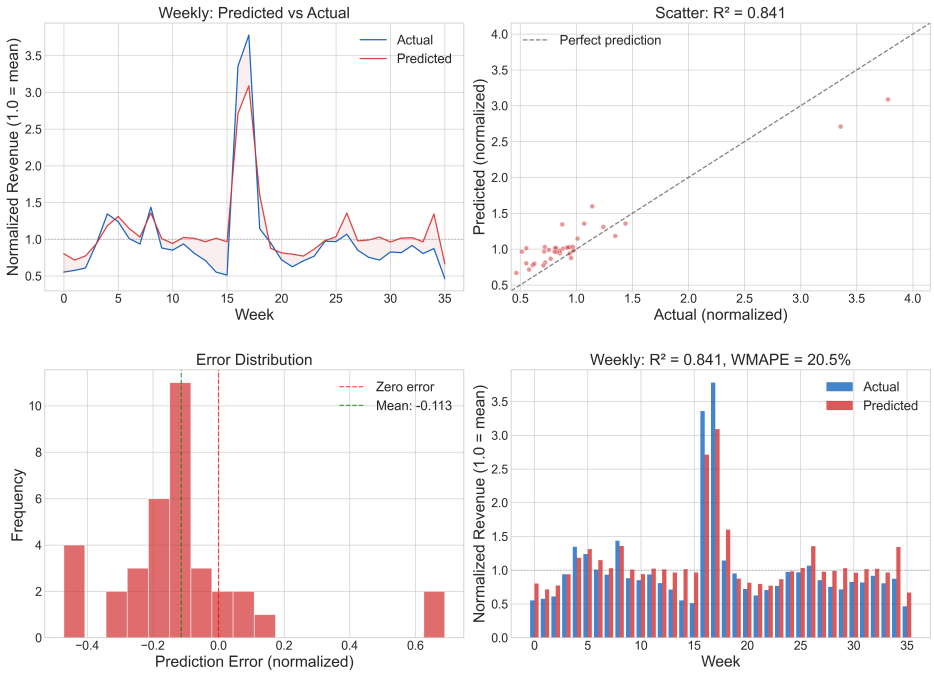


Figure 5.1: Neural model prediction diagnostics for Merchant 3 (test set).

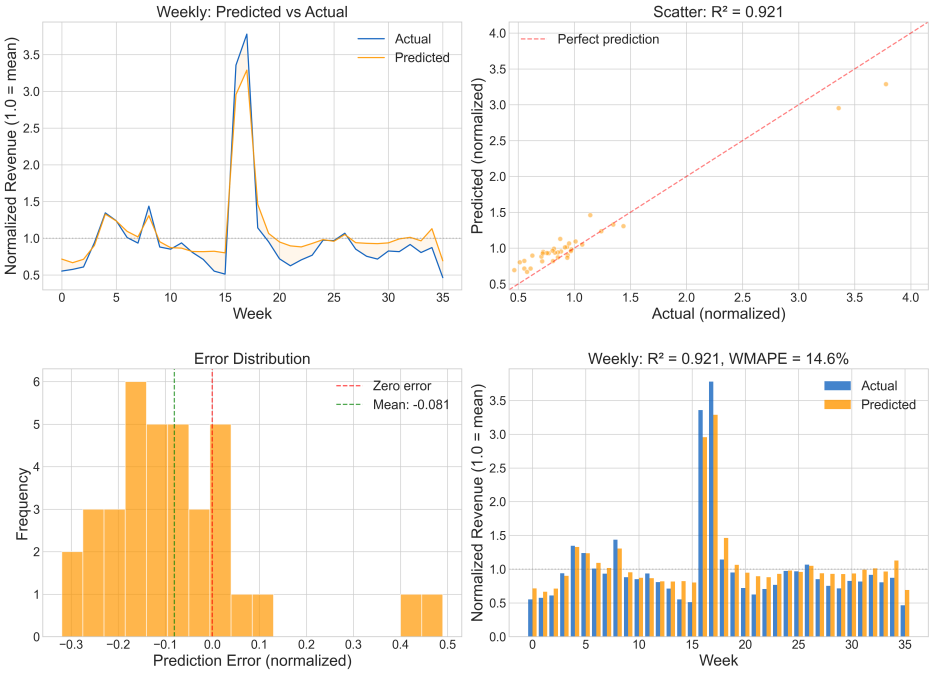


Figure 5.2: Bayesian MMM prediction diagnostics for Merchant 3 (test set).

Both models capture the overall demand trend, but exhibit different failure modes. The large demand spike visible around week 15 corresponds to the Black Friday promotional period, a recurring seasonal event that can multiply weekly revenue by 3–5×. Both models underestimate this peak, but the neural model does so more severely, which explains a substantial portion of the R^2 gap. Accurately capturing such extreme seasonality is a well-known challenge and underscores the importance of the baseline component's seasonal features. Outside of this event, the two models produce comparable predictions. The neural model's error distribution shows a slight negative bias (mean -0.113), indicating a systematic tendency to underpredict; the Bayesian MMM's distribution is more centered (mean -0.081) with slightly lower variance.

5.3 Temporal Dynamics

5.3.1 Learned Temporal Profiles

The neural model's per-channel temporal attention mechanism, described in Section 4.1.3, produces channel-specific temporal response functions

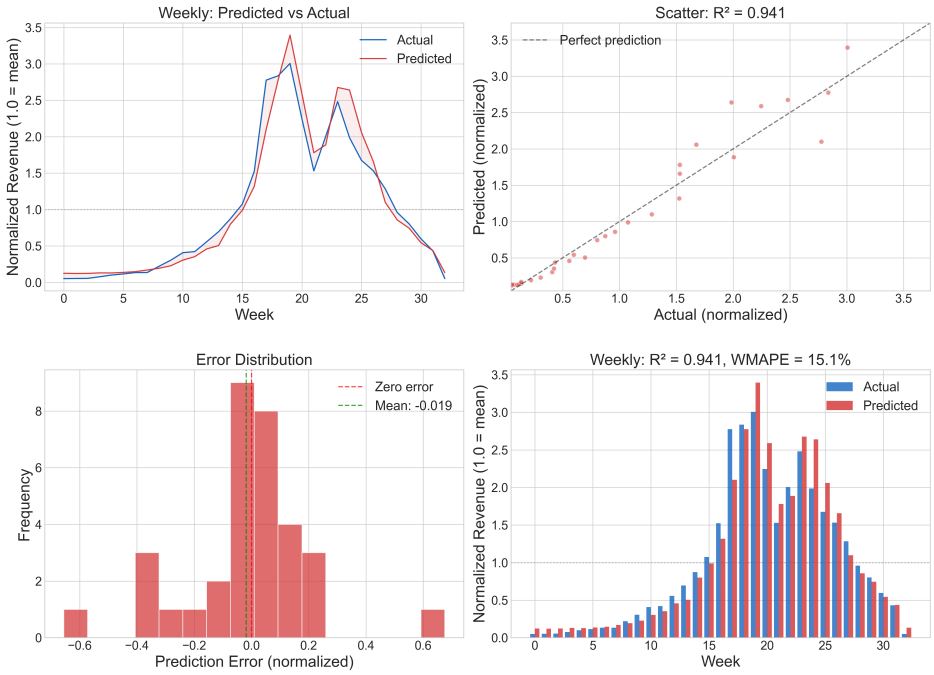


Figure 5.3: Neural model prediction diagnostics for Merchant 1 (test set).

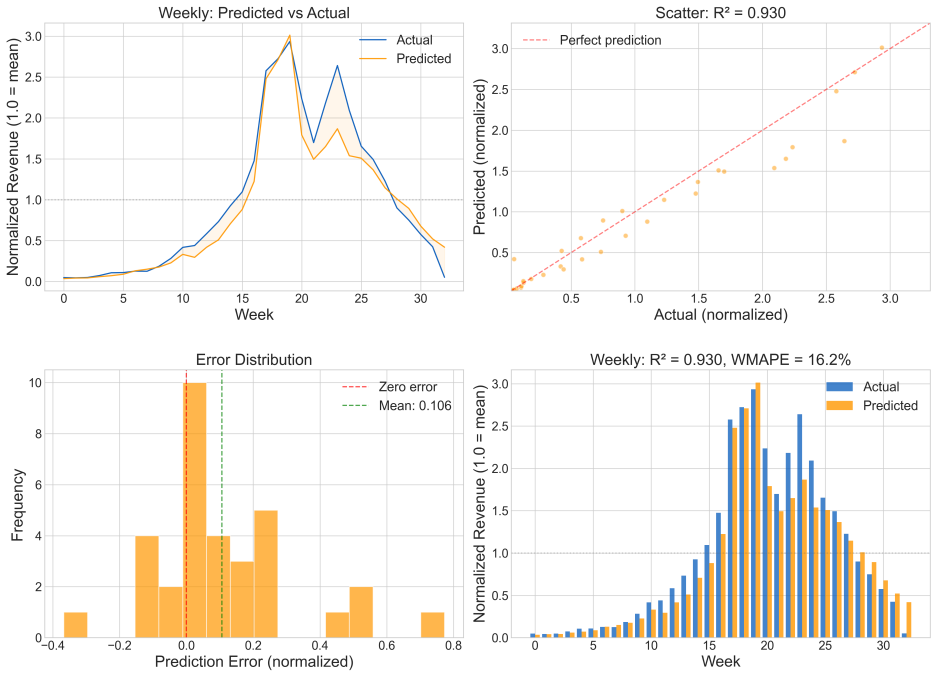


Figure 5.4: Bayesian MMM prediction diagnostics for Merchant 1 (test set).

learned directly from the data. Figure 5.5 shows the effective cumulative adstock curves for a representative merchant, obtained as the scale-weighted combination of per-channel attention profiles across all temporal heads.

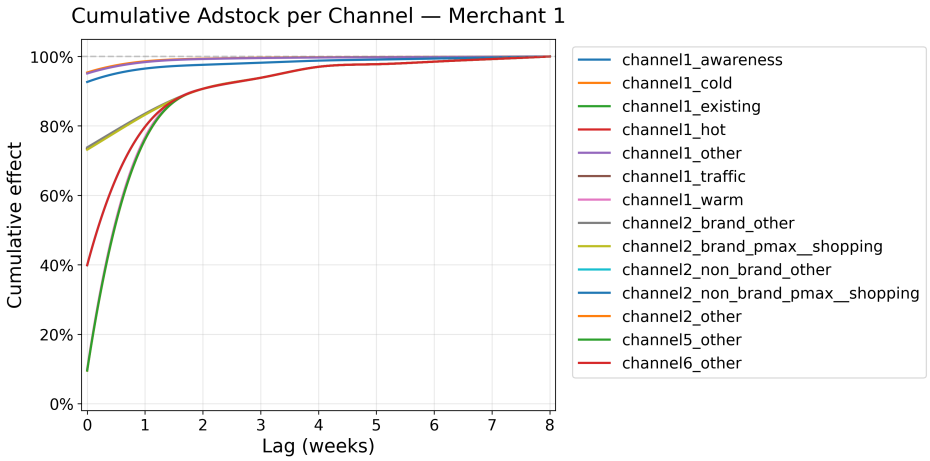


Figure 5.5: Neural model: cumulative adstock per channel for a representative merchant. Each curve shows the fraction of the total temporal effect realized by a given lag. Channels exhibit qualitatively distinct dynamics, from steep immediate saturation to gradual delayed accumulation. Results for all merchants are reported in Appendix B.

The learned profiles reveal substantial heterogeneity across channels. Lower-funnel channels (e.g., branded search, retargeting) show steep initial curves, with most of the effect concentrated in the first 1–2 weeks, consistent with the expectation that conversion-oriented campaigns produce rapid, short-lived responses. Upper-funnel channels (e.g., awareness campaigns, video platforms) tend to exhibit more distributed profiles, with the effect accumulating over 3–5 weeks, although the distinction is not always as sharp as domain expectations would suggest.

A notable feature is that many channels start at a non-zero cumulative effect at lag zero, meaning the model attributes a portion of the response to the same week as the spend. While this is plausible for direct-response channels (e.g., branded search), the same-week effect tends to be disproportionately large for some upper-funnel channels where an instantaneous conversion would not be expected given the typical adstock profile. Without domain priors constraining the minimum lag, the purely data-driven approach risks absorbing contemporaneous spend–revenue correlations into the temporal profile, potentially mixing spurious co-movements with genuine marketing effects.

5.3.2 Bayesian MMM Adstock

Figure 5.6 shows the equivalent cumulative adstock curves produced by the Bayesian MMM for the same merchant.

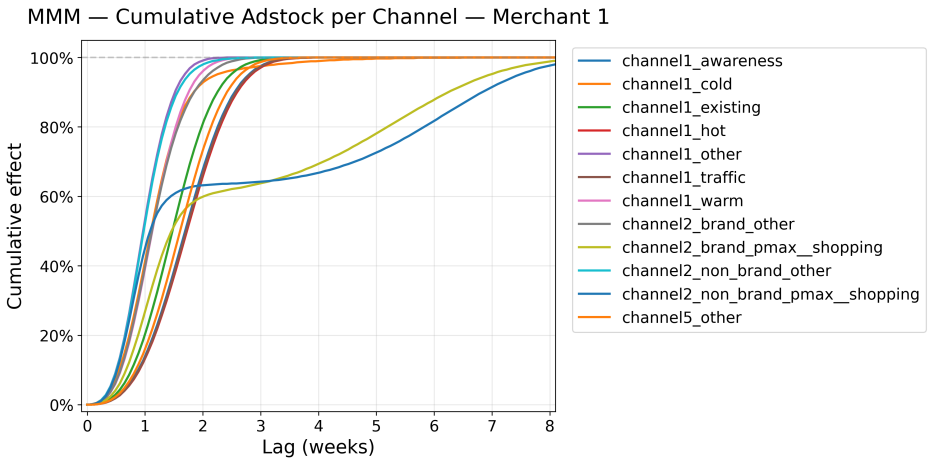


Figure 5.6: Bayesian MMM: cumulative adstock per channel for a representative merchant. All curves follow parametric S-shaped forms (Weibull decay), starting from zero at lag zero.

The parametric adstock curves show a fundamentally different structure. All channels start at zero cumulative effect at lag zero and follow smooth S-shaped trajectories, a direct consequence of the chosen functional form. While the curves do differentiate between channels (some saturate within 2 weeks, others extend to 6–8 weeks), the overall diversity is more constrained than in the neural model.

The key structural difference between the two approaches is the source of temporal information. The Bayesian MMM's adstock shapes are determined by the parametric functional form (geometric or Weibull decay) whose parameters are estimated from the data but whose shape class is fixed a priori. The neural model learns the entire temporal profile non-parametrically through per-channel MLPs, allowing it to represent arbitrary response shapes including non-monotonic patterns and instantaneous effects that the parametric form cannot express.

5.3.3 Comparison of Temporal Profiles

Figure 5.7 provides a direct comparison between the neural and Bayesian adstock curves for the three channels with the largest temporal variation.

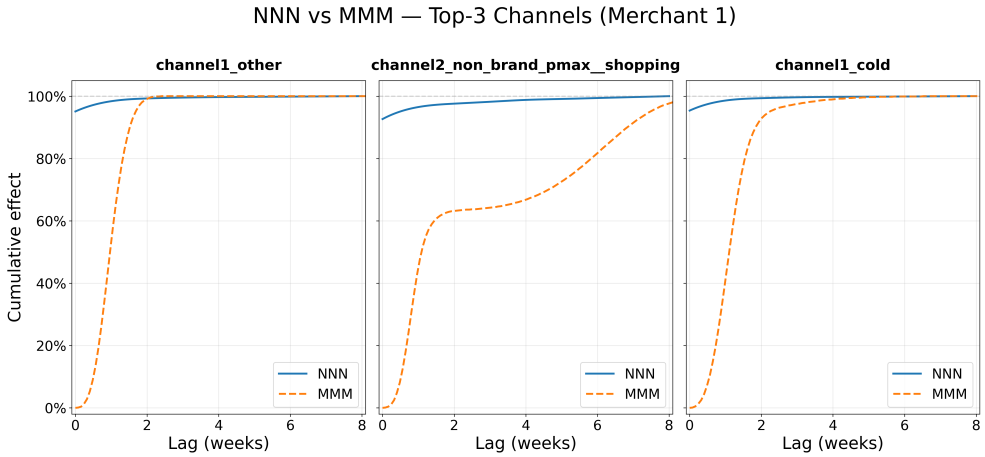


Figure 5.7: NNN vs MMM cumulative adstock for the top-3 channels by total spend (representative merchant). Solid blue: neural model; dashed orange: Bayesian MMM. The neural model saturates faster, capturing a larger instantaneous effect; the Bayesian MMM distributes the response more gradually over time.

For the same channels, the two models produce qualitatively different temporal profiles. The neural model consistently shows higher initial effects and faster saturation, while the Bayesian MMM distributes the response more gradually over longer horizons. This pattern is visible across all three channels: the neural curves reach 80–100% of the total effect within 1–2 weeks, while the MMM curves may take 4–6 weeks to reach the same level.

Neither set of profiles can be declared "correct" in the absence of ground-truth temporal response data. The Bayesian MMM's parametric adstock allows practitioners to embed domain knowledge, such as expected decay rates informed by prior studies, into the temporal structure, which can be valuable when data alone is insufficient. However, the chosen functional form constrains the set of representable shapes, and when the true decay is faster or more irregular than the parametric family can express, the resulting profiles may spread effects over artificially long horizons. The neural model trades this domain-informed structure for flexibility: it can represent arbitrary response shapes but, as discussed above, risks overfitting to contemporaneous correlations when the signal is weak.

5.4 Cross-Channel Interactions

The cross-channel interaction module described in Section 4.1.4 has no counterpart in the Bayesian MMM, which assumes additive, independent channel effects. This section examines whether the learned interaction patterns are interpretable and consistent with marketing domain knowledge.

The Weibull-parametrized interaction kernels (Figures 4.6–4.7 in Chapter 4) reveal structured spillover patterns between specific channel pairs. Two consistent patterns emerge across merchants:

1. **Funnel conversion effects**: upper- and mid-funnel channels show positive spillovers toward lower-funnel channels, reflecting the expected demand propagation from awareness and consideration to conversion. Spend on awareness or prospecting campaigns generates interest that materializes as purchases downstream.

2. **Within-platform vs cross-platform**: these funnel effects appear both within the same advertising platform and across different platforms. Within-platform interactions tend to peak earlier (0–1 weeks), while cross-platform interactions exhibit longer delays (2–4 weeks) and wider profiles, consistent with the slower path from initial exposure on one platform to conversion on another.

Figure 5.8 summarizes the top cross-channel pairs ranked by temporal variation for a representative merchant. Figures 5.9 and 5.10 isolate two pairs that exemplify the within-platform and cross-platform variants of these funnel conversion effects.

Although the Weibull kernel initialization and the diagonal prior described in Section 4.1.4 do bias the model toward plausible interaction shapes, the specific patterns (which pairs interact, with what delay, and at what magnitude) are learned from the data. Given the limited number of weekly observations ($\sim$140–240 per merchant), these results should be interpreted cautiously: the model may not have enough statistical power to reliably distinguish all pairwise interactions from noise, particularly for low-spend channel pairs. Nevertheless, the consistency of the funnel-aligned patterns across merchants suggests that the cross-channel module captures at least the dominant dynamics.

The Bayesian MMM's independence assumption precludes this type of analysis entirely. Any cross-channel demand propagation is either absorbed by the baseline component or misattributed to the originating channel. The neural

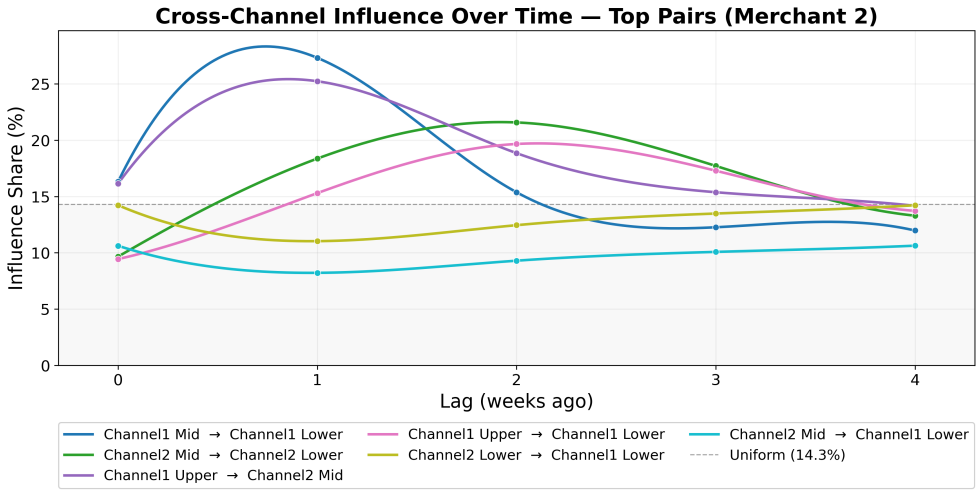


Figure 5.8: Cross-channel influence over time for the top pairs by temporal variation (representative merchant). Each curve shows the influence share that a source channel exerts on a target channel as a function of lag. The dashed gray line marks the uniform baseline.

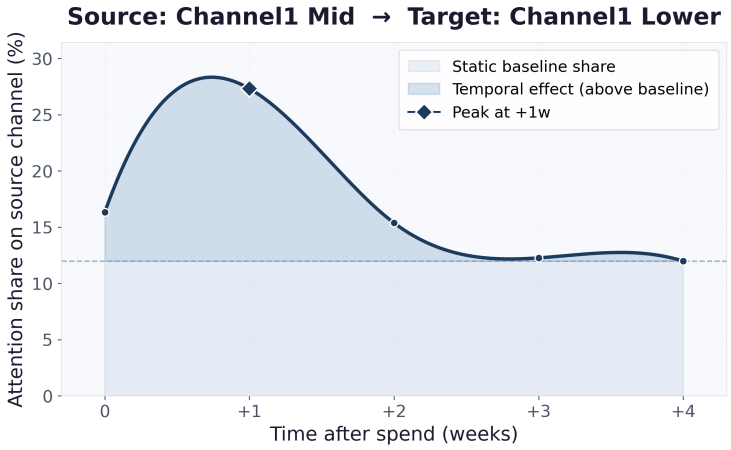


Figure 5.9: Short-delay funnel propagation: mid-funnel → lower-funnel. The impulse response peaks at +1 week and decays rapidly, with influence returning to the baseline within 3–4 weeks.

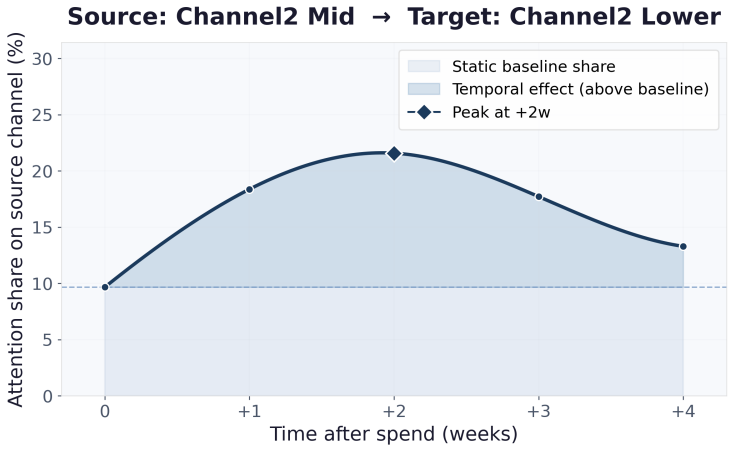


Figure 5.10: Broader delayed interaction: mid-funnel → lower-funnel with a peak at +2 weeks and a wider profile, indicating a slower demand pipeline.

model's explicit cross-channel mechanism provides a structural advantage for understanding how marketing investments interact across the media mix, even if the limited data constrains the precision of the estimates.

5.5 Attribution and Decomposition

The counterfactual leave-one-out procedure described in Section 4.5 decomposes the predicted outcome into per-channel media contributions and an organic baseline. This section compares the resulting decomposition with the channel-level attribution produced by the Bayesian MMM for Merchant 1, the dataset with the richest channel mix (14 channels).

5.5.1 Channel Contribution Breakdown

Figures 5.11 and 5.12 show the temporal decomposition of predicted Net Gross Profit for the neural model and the Bayesian MMM, respectively. Both models identify the same dominant channels and produce a broadly consistent ranking of media contributions, indicating that the neural model recovers a plausible decomposition despite learning the channel weights entirely from data without parametric priors.

The two decompositions differ in two important respects. First, the Bayesian MMM attributes a larger share to the organic baseline (approximately 52% of the total prediction, versus 37% for the neural model). The Bayesian model's parametric adstock and saturation priors constrain the

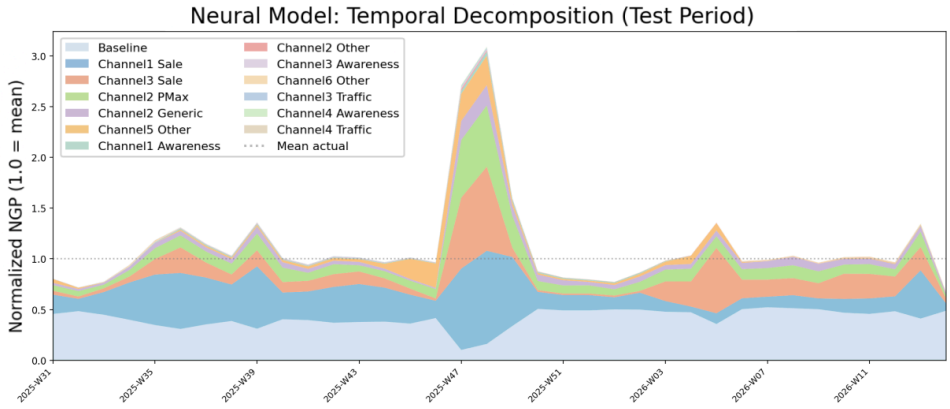


Figure 5.11: Neural model: temporal decomposition of predicted Net Gross Profit into baseline and per-channel media contributions over the test period (Merchant 1).

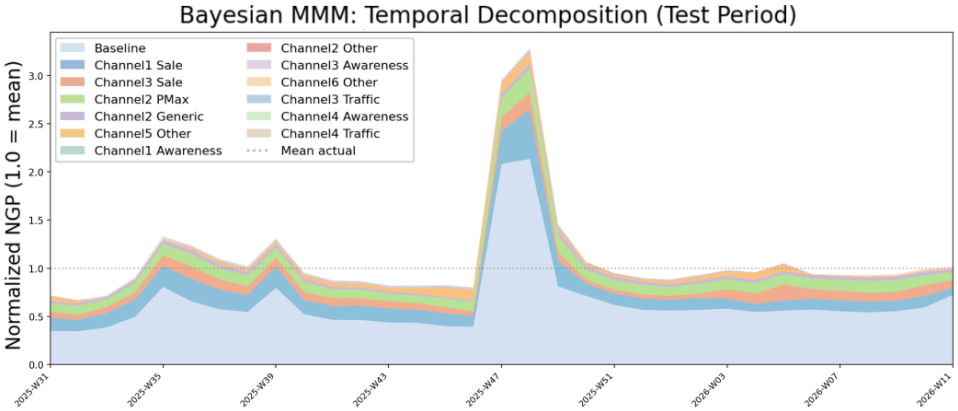


Figure 5.12: Bayesian MMM: temporal decomposition of attributed Net Gross Profit into baseline and per-channel media contributions over the same test period (Merchant 1).

magnitude of media effects, channeling residual variation into the baseline; the neural model, lacking analogous constraints, distributes more of the predicted output across media channels. Second, the neural decomposition exhibits higher week-to-week volatility. The spike around 2025-W47 is visible in both models, but the neural model amplifies it substantially (peaking above 3× the mean), with individual channel contributions fluctuating sharply. This volatility reflects the model's greater representational freedom in a limited-data regime: without structural constraints anchoring the attribution, weekly correlations between spend and revenue can produce transient spikes that do not necessarily reflect stable causal effects.

5.5.2 Channel Attribution Comparison

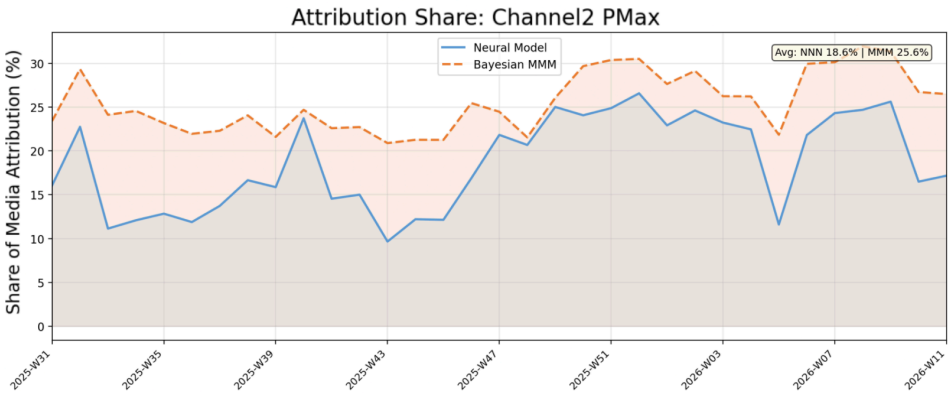


Figure 5.13: Weekly attribution share for Channel2 PMax (mid-funnel performance shopping, Merchant 1). Solid blue: neural model; dashed orange: Bayesian MMM.

Figures 5.13–5.14 isolate two representative channels to compare the weekly attribution share assigned by each model. For Channel2 PMax, the models diverge substantially: the Bayesian MMM attributes roughly 23% of the total media effect to this channel, while the neural model assigns only 14%, suggesting that the unconstrained model redistributes part of this channel's contribution elsewhere. For Channel1 Sale, both models agree that it is the single largest contributor (NNN ∼41%, MMM ∼35%), but the neural estimate again shows higher temporal variance.

The pattern across the two channels is consistent: the neural model's non-parametric decomposition captures the broad structure of the attribution but introduces additional volatility that the Bayesian MMM's parametric

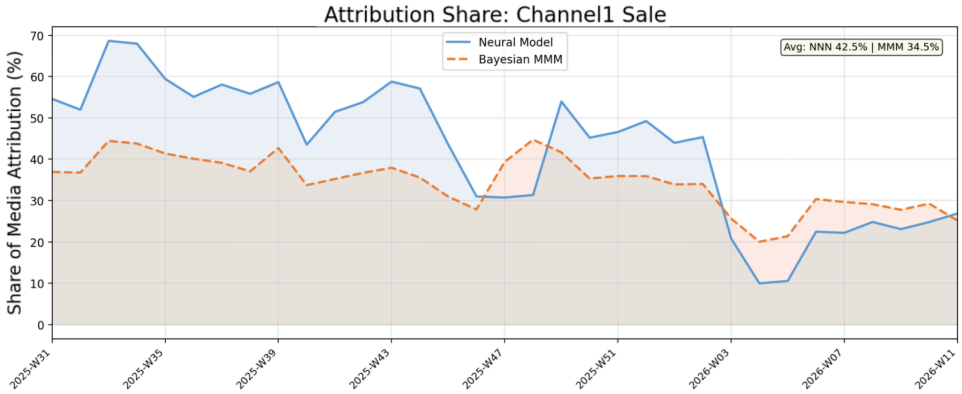


Figure 5.14: Weekly attribution share for Channel1 Sale (lower-funnel social conversion, Merchant 1). Solid blue: neural model; dashed orange: Bayesian MMM.

constraints suppress. Whether this volatility reflects genuine week-to-week dynamics or overfitting to noisy correlations cannot be determined without experimental ground truth. In this data regime, the parametric approach trades representational flexibility for stability, a trade-off that appears favorable when the decomposition must be reliable. In practice, unstable attribution estimates propagate directly into the return-on-ad-spend metrics that drive budget allocation decisions, making stability a prerequisite for actionable insights.

5.6 Reliability Analysis

The reliability of the reported results depends on three factors: training stability across random seeds, sensitivity to hyperparameter choices, and reproducibility of the main qualitative findings.

Training stability is addressed through the early stopping protocol and learning rate scheduling described in Section 4.3. The three-phase Bayesian hyperparameter optimization (Section 4.4) reduces the risk that results depend on a lucky parameter configuration.

To quantify seed sensitivity, the model was trained three times on Merchant 1 with identical hyperparameters but different random seeds. Table 5.3 reports the results.

The R^2 values span a range of 0.012 and WMAPE varies by approximately one percentage point across seeds. This level of variation is modest relative to the absolute performance and does not affect the qualitative conclusions. The main patterns reported in this chapter (channel-heterogeneous temporal

Table 5.3: Seed variance study for Merchant 1 (3 independent runs, same hyperparameters).

	Seed 1	**Seed 2**	**Seed 3**
R^2	0.929	0.941	0.933
WMAPE	15.9%	16.1%	15.2%

profiles, structured cross-channel interactions, and the accuracy gap relative to the Bayesian MMM) were consistently observed across all three runs.

5.7 Validity Analysis

The following considerations should be kept in mind when interpreting the results.

Internal validity. As with all marketing mix analyses, the evaluation relies on observational data. While this precludes strict causal claims, both models operate under the same assumption, that variation in spend is informative about its effect on outcomes, and the comparison between the two is conducted on identical data. The absence of ground-truth attribution weights means that the structural analysis (Sections 5.3–5.4) evaluates plausibility and consistency with domain expectations rather than absolute correctness.

External validity. The three evaluation merchants span different product categories, price points, and channel mixes within the e-commerce domain, providing reasonable diversity for the target use case. The architecture is designed for and evaluated on e-commerce businesses with digital advertising budgets; generalization to fundamentally different settings (e.g., subscription models, B2B) would require further validation.

Construct validity. The comparison between the neural model and the Bayesian MMM is asymmetric in maturity: the Bayesian system has been iteratively refined over production deployment cycles, while the neural model is evaluated as a research prototype. The accuracy gap reported in Section 5.2 reflects this maturity difference in addition to any inherent architectural differences.

Chapter 6

Discussion

This chapter interprets the empirical findings reported in Chapter 5 at a higher level of abstraction. It explicitly addresses the research question formulated in Chapter 1, discusses the trade-offs between the two modeling paradigms, and connects the results to the broader literature. Detailed limitations and directions for future work are deferred to Chapter 7.

6.1 Answering the Research Question

The thesis asked to what extent attention-based neural Marketing Mix Models can achieve predictive performance comparable to Bayesian MMMs while enabling richer structural analyses of temporal and cross-channel marketing dynamics under realistic data constraints. The results suggest a qualified answer: the neural model does not consistently match the production Bayesian MMM as a predictive tool, but it reaches a level of accuracy that is sufficient to support meaningful structural analyses, while exposing temporal and cross-channel patterns that the parametric framework does not represent directly.

In predictive performance, the Bayesian MMM remains stronger overall, achieving higher R^2 and lower WMAPE in most of the available comparison (Section 5.2.1). This gap reflects several factors: the Bayesian system benefits from domain-informed priors that regularize the model effectively with limited data, from a longer optimization history, and from a mature infrastructure for data preprocessing and feature engineering (Section 5.2). At the same time, the neural model achieves comparable performance in some settings and captures enough variance to make its internal temporal and cross-channel representations analytically informative.

Where the neural model distinguishes itself is in the structural representa-

tions it produces. The temporal attention mechanism learns channel-specific response profiles that are substantially more diverse than the parametric adstock curves of the Bayesian MMM (Section 5.3.3), capturing heterogeneous decay patterns without constraining them to a predefined functional family. At the same time, the absence of parametric constraints introduces a risk: the tendency of several channels to exhibit disproportionately large same-week effects suggests that the model may absorb contemporaneous correlations into the temporal profile when the signal is weak. The cross-channel interaction module represents the most distinctive contribution. By learning pairwise Weibull-parametrized impulse response functions, the model captures spillover dynamics (funnel conversion effects, within-platform and cross-platform propagation) that the Bayesian MMM's independence assumption cannot represent (Section 5.4). These patterns are consistent with marketing domain expectations, although the limited sample size constrains the precision of the estimates.

The attribution decomposition (Section 5.5) reinforces this conclusion: both models identify the same dominant channels, but the neural model's unconstrained decomposition exhibits higher temporal volatility, while the Bayesian MMM's parametric constraints produce more stable estimates suitable for budget allocation.

Taken together, the evidence indicates that attention-based architectures can operate meaningfully in the low-sample, high-noise regime typical of aggregated marketing data. Their predictive performance is not consistently superior to, nor fully interchangeable with, the Bayesian MMM baseline. However, the neural model provides richer structural analyses of temporal heterogeneity, cross-channel dynamics, and fine-grained attribution, making the two approaches complementary rather than competing.

6.2 Flexibility vs Domain Knowledge

The central tension examined in this thesis is the trade-off between data-driven flexibility and domain-informed structure.

The Bayesian MMM embeds marketing knowledge directly into its functional form: parametric adstock functions constrain the temporal response to smooth, monotonic shapes; Michaelis-Menten saturation encodes diminishing returns; and Bayesian priors regularize the parameter space based on prior studies and expert judgment. These structural assumptions make the model robust with limited data and interpretable by design, but they also limit its representational capacity. Effects that do not conform to

the assumed functional family (non-monotonic temporal responses, cross-channel spillovers, time-varying dynamics) are either absorbed into the baseline or left unmodeled.

The neural model takes a more flexible approach: it learns temporal profiles, channel interactions, and response shapes directly from the data through attention mechanisms and parametric kernels. This flexibility enables the discovery of patterns that the parametric framework cannot express, as demonstrated by the heterogeneous temporal profiles and the structured cross-channel interactions reported in Chapter 5. However, flexibility comes at a cost: without domain priors, the model requires explicit regularization (temporal diversity loss, L1 penalties) to avoid degenerate solutions, and it remains susceptible to overfitting contemporaneous correlations, particularly for channels with weak signals.

The results suggest that continuing to explore the neural direction is justified, not because it replaces the Bayesian MMM, but because it exposes structural information that complements the predictive and attribution outputs of the production baseline. The temporal profiles and cross-channel interactions learned by the model capture aspects of marketing dynamics that are not accessible through the parametric framework, and these representations could potentially inform and enrich existing models as well. The two paradigms capture complementary aspects of the underlying system: the Bayesian MMM provides robust aggregate estimates grounded in domain knowledge, while the neural model offers a lens for structural discovery and fine-grained temporal analysis.

6.3 Practical Implications

The findings have implications for practitioners operating in complex digital marketing ecosystems with many channels and potential cross-platform interactions.

The cross-channel interaction module provides information that is not available from independent-channel models: which channel pairs exhibit spillover effects, with what delay, and at what relative magnitude. For media planners, this type of structural insight can inform sequencing decisions (e.g., timing upper-funnel campaigns to precede lower-funnel pushes by the estimated lag) and help identify synergies that aggregate budget allocation models would miss. The funnel conversion patterns observed across merchants (Section 5.4) are consistent with established marketing theory but are here quantified from observational data rather than assumed a priori.

The neural model is not ready to replace the Bayesian MMM in production settings. The accuracy gap, the sensitivity to hyperparameter configuration, and the interpretability challenges associated with learned representations all argue for caution. A more realistic deployment scenario positions the two models as complementary tools: the Bayesian MMM for robust budget allocation and performance forecasting, and the neural model for exploratory analysis of temporal and cross-channel dynamics. This is consistent with the industrial context described in Section 1.2.1, where the existing MMM infrastructure provides a stable foundation that the neural approach can augment rather than replace.

6.4 Relation to Prior Work

The results can be situated within the broader landscape of marketing measurement research.

The NNN framework proposed by Mulc et al. (2025) demonstrated that attention-based architectures can capture temporal and cross-channel effects in marketing data. The present work extends this investigation to settings with 9–14 marketing channels and limited weekly observations ($\sim$140–240), compared to the simpler configurations evaluated in the original paper. The core architectural principles (per-channel temporal attention, cross-channel interaction kernels, counterfactual attribution) transfer to this more complex setting, but the limited data availability introduces challenges that were not prominent in the original evaluation, including the lag-zero overfitting tendency and the need for diversity regularization.

Chan and Perry (2017) documented key challenges of traditional MMMs, including sensitivity to parametric assumptions and difficulties in handling correlated channels. The temporal diversity observed in the neural model's learned profiles provides indirect support for their observation that fixed functional forms may not capture the full heterogeneity of marketing effects. At the same time, the accuracy advantage of the Bayesian MMM in our comparison confirms that domain-informed priors remain valuable when data are scarce.

The time-varying coefficient model proposed by Ng et al. (2021) represents an intermediate approach: it relaxes the assumption of time-invariant parameters within a Bayesian framework. The neural model's temporal profiles, which are learned non-parametrically, can be seen as a more flexible version of the same idea, allowing the temporal structure to vary not only over time but also across channels and temporal scales, at the cost of

reduced interpretability and weaker regularization.

Chapter 7

Conclusions and Future work

This chapter summarizes the contributions of the thesis, identifies the main limitations, outlines directions for future research, and offers reflections on the broader implications of the work.

7.1 Conclusions

This thesis investigated the extent to which attention-based neural Marketing Mix Models can achieve predictive performance comparable to Bayesian MMMs while enabling richer structural analyses of temporal and cross-channel marketing dynamics using aggregated time-series data. The work was structured around three goals, which are revisited below.

The primary contribution is scientific and engineering in nature. The thesis provides empirical evidence on how attention-based models behave in the low-sample, high-noise regime characteristic of aggregated marketing data. Three findings stand out. First, the per-channel temporal attention mechanism produces response profiles that are substantially more heterogeneous than those obtainable through parametric adstock functions, demonstrating that the temporal structure of marketing effects is richer than what classical functional forms can represent. Second, the cross-channel interaction module, which has no counterpart in traditional Marketing Mix Models, captures spillover patterns that are consistent with established funnel theory, quantifying inter-channel dynamics from observational data rather than assuming them a priori. Third, the lag-zero overfitting tendency identified in the learned temporal profiles highlights a fundamental trade-off in purely data-driven approaches: the same flexibility that enables the discovery of unexpected patterns also risks absorbing spurious contemporaneous correlations when domain priors

are absent. This finding contributes to the broader understanding of the tension between flexibility and inductive bias in deep learning applied to small-sample time series.

From an industrial perspective, the empirical comparison across three real-world e-commerce merchants confirms that the production Bayesian MMM remains more accurate as a predictive tool overall. However, the neural model achieves comparable predictive performance in some settings and provides structural analyses (temporal response heterogeneity, cross-channel spillover quantification) that are not obtainable from the independent-channel framework of the Bayesian MMM. The two approaches are therefore complementary rather than competing.

Finally, the educational goal of demonstrating methodological rigor has been addressed through a reproducible experimental pipeline, a structured comparative evaluation, and a critical analysis of both the capabilities and limitations of each approach.

Taken together, the results show that pursuing the neural direction for marketing measurement is justified not only by predictive performance, but by the structural representations it produces. The insights gained, particularly regarding temporal heterogeneity and cross-channel dynamics, can enrich the understanding of marketing systems while complementing the robustness of established Bayesian MMMs.

7.2 Limitations

Beyond the validity considerations discussed in Section 5.7, several concrete limitations constrain the scope and interpretation of this work.

The available sample sizes ($\sim$140–240 weekly observations per merchant) are sufficient to identify dominant temporal and cross-channel patterns but limit the statistical power for estimating weaker pairwise interactions. The cross-channel module may not reliably distinguish subtle spillover effects from noise, particularly for low-spend channel pairs.

No ground-truth attribution data are available. In the absence of controlled experiments (A/B tests, geo-experiments), the plausibility of the learned temporal profiles and interaction patterns is assessed qualitatively, by consistency with domain expectations, rather than quantitatively validated against causal benchmarks.

Finally, the current evaluation is limited to channels represented by aggregate spend and volume data. The framework has not been tested with channels for which richer signals (such as creative content embeddings or

search query representations) are available, a setting that Mulc et al. (2025) identify as central to the full NNN framework.

7.3 Future work

The following directions extend the present work and are in part shared with those identified by Mulc et al. (2025) for the original NNN framework.

Validation through live experiments. As emphasized by Mulc et al. (2025), retrospective evaluation on observational data cannot fully establish the validity of attribution estimates. Comparing model outputs against results from A/B tests or geo-experiments is the most critical next step for establishing practical credibility.

Minimum-lag constraints. The lag-zero overfitting identified in this work could be mitigated by introducing a learnable or fixed minimum-lag parameter that prevents the model from concentrating temporal effects at lag zero. Alternatively, soft parametric priors on the temporal profile shape could regularize the learned curves without fully constraining them to a predefined functional family.

Architectural extensions. Cross-geographical attention and advanced embedding aggregation strategies were explicitly excluded from the scope of this thesis (Chapter 1, Delimitations). Now that the base architecture has been validated, these extensions represent a natural next step. Mulc et al. (2025) also suggest multi-head attention to allow different heads to specialize on distinct temporal patterns or channel interactions.

Cost integration and budget optimization. The current framework operates on spend volumes. Integrating per-channel cost data would enable direct computation of Return on Ad Spend (ROAS) and open the path toward budget optimization based on the model's structural outputs.

Hybrid approaches. A promising direction is to explore architectures that combine the flexibility of neural representations with structural priors inspired by Bayesian models, for instance soft parametric constraints on temporal profile shapes or informative priors on cross-channel weights. The goal would be to preserve the neural model's capacity for structural discovery

while reducing its sensitivity to spurious correlations and its dependence on explicit regularization.

7.4 Use of Generative AI

Generative AI tools were used during the preparation of this thesis to improve the clarity, readability, and professional quality of the written language. All intellectual contributions, including the research question, the architectural design, the experimental methodology, the analysis, and the conclusions, are entirely the author's own work. At no point was AI used to generate ideas, produce novel content, or substitute independent scientific reasoning.

7.5 Ethical and Sustainability Considerations

The analysis relies exclusively on aggregated marketing spend and revenue data, with no user-level tracking or personally identifiable information. This is consistent with GDPR requirements and avoids the privacy concerns associated with individual-level attribution methods.

From a computational sustainability perspective, the neural model's training requirements are modest. With the architectural and optimization choices described in Chapter 4 (early stopping, phased Bayesian hyperparameter search, and a contained model size), the training process does not require specialized hardware, making the approach accessible and its environmental footprint limited.

A less obvious consideration concerns the downstream effects of attribution models on marketing practice. Models that inform budget allocation decisions can create self-reinforcing feedback loops: channels with rich historical data produce more confident estimates, attracting further investment, while newer or smaller channels remain underrepresented. Practitioners should be aware of this dynamic and complement model-driven allocation with deliberate experimentation on underexplored channels.

References

Akiba, T., Sano, S., Yanase, T., Ohta, T., and Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 2623–2631.

Broadbent, S. (1979). One way tv advertisements work. *Journal of the Market Research Society*, 21(3).

Caruana, R. (1997). Multitask learning. *Machine Learning*, 28(1):41–75.

Chan, D. and Perry, M. (2017). Challenges and opportunities in media mix modeling. Technical report, Google, Inc.

Dew, R., Padilla, N., and Shchetkina, A. (2024). Your mmm is broken: Identification of nonlinear and time-varying effects in marketing mix models. *arXiv preprint arXiv:2408.07678*.

Gong, C., Yao, D., Zhang, L., Chen, S., Li, W., Su, Y., and Bi, J. (2024). Causalmmm: Learning causal structure for marketing mix modeling. In *Proceedings of the 17th ACM International Conference on Web Search and Data Mining*, WSDM '24, page 238–246, New York, NY, USA. Association for Computing Machinery.

Goodfellow, I., Bengio, Y., and Courville, A. (2016). *Deep Learning*. MIT Press.

Gujar, P., Paliwal, G., Panyam, S., and Kewalramani, C. (2024). The evolution of ads marketing mix modeling: From regression models to ai-powered planning for smbs. In *IEEE Technology and Engineering Management Society (TEMSCON LATAM)*, pages 1–6.

Guo, C. and Berkhahn, F. (2016). Entity embeddings of categorical variables. *arXiv preprint arXiv:1604.06737*.

Guzman, W. (2024). *The Good, the Efficient and the Inductive Biases:: Exploring Efficiency in Deep Learning Through the Use of Inductive Biases*. PhD thesis, VU E-Publishing.

Hanssens, D. M., Parsons, L. J., and Schultz, R. L. (2001). *Market Response Models: Econometric and Time Series Analysis*. Kluwer Academic Publishers, 2nd edition.

He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778.

Hoffman, M. D. and Gelman, A. (2014). The No-U-Turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15(47):1593–1623.

Jin, Y., Wang, Y., Sun, Y., Chan, D., and Koehler, J. (2017). Bayesian methods for media mix modeling with carryover and shape effects. Technical report, Google Inc.

Kingma, D. P. and Ba, J. (2015). Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*.

Lim, B., Arık, S. Ö., Loeff, N., and Pfister, T. (2021). Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4):1748–1764.

Mei, H. and Eisner, J. (2017). The neural hawkes process: A neurally self-modulating multivariate point process. In *Advances in Neural Information Processing Systems*, volume 30.

Michaelis, L. and Menten, M. L. (1913). Die Kinetik der Invertinwirkung. *Biochemische Zeitschrift*, 49:333–369.

Miller, K. M., Lukic, K., and Skiera, B. (2026). The impact of the general data protection regulation (gdpr) on online tracking. *International Journal of Research in Marketing*, 43(1):48–70.

Mulc, T., Anderson, M., Cubre, P., Zhang, H., Liu, I., and Kumar, S. (2025). Nnn: Next-generation neural networks for marketing measurement.

Naik, P. A. and Raman, K. (2003). Understanding the impact of synergy in multimedia communications. *Journal of Marketing Research*, 40(4):375–388.

Ng, E., Wang, Z., and Dai, A. (2021). Bayesian time varying coefficient model with applications to marketing mix modeling. In *Proceedings of the AdKDD Workshop at the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*.

Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Köpf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., Bai, J., and Chintala, S. (2019). PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32, pages 8024–8035.

Prechelt, L. (2012). Early stopping — but when? In *Neural Networks: Tricks of the Trade*, volume 7700 of *Lecture Notes in Computer Science*, pages 53–67. Springer, 2nd edition.

Runge, J., Skokan, I., Zhou, G., and Pauwels, K. (2024). Robyn: Semi-automated marketing mix modeling from meta marketing science. *arXiv preprint arXiv:2403.14674*.

Sims, C. A. (1980). Macroeconomics and reality. *Econometrica*, 48(1):1–48.

Snoek, J., Larochelle, H., and Adams, R. P. (2012). Practical Bayesian optimization of machine learning algorithms. In *Advances in Neural Information Processing Systems*, volume 25, pages 2951–2959.

Spiliotis, E., Makridakis, S., and Semenoglou, A. A. (2022). Comparison of statistical and machine learning methods for daily sku demand forecasting. *Operational Research International Journal*, 22:3037–3061.

Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., and Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(56):1929–1958.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30.

Vaver, J. and Zhang, S. S.-H. (2017). Introduction to the aggregate marketing system simulator. Technical report, Google, Inc.

Weibull, W. (1951). A statistical distribution function of wide applicability. *Journal of Applied Mechanics*, 18(3):293–297.

Zhang, Y., Wurm, M., Li, E., Wakim, A., Kelly, J., Price, B., and Liu, Y. (2024). Media mix model calibration with bayesian priors. Technical report, Google.

Zhou, H., Zhang, S., Peng, J., Zhang, S., Li, J., Xiong, H., and Zhang, W. (2021). Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 11106–11115.

Appendix A

Hyperparameter Optimization Details

This appendix reports the optimization progress and parameter distributions from the phased Bayesian hyperparameter optimization described in Section 4.4. All plots refer to a representative merchant (Merchant 1). Each phase fixes the parameters selected in the previous phase and searches a new subset, progressively narrowing the configuration space.

A.1 Phase 1: Stabilization

The first phase targets the parameters most critical for training stability: learning rate, L_1 regularization strength, and baseline weight. Figure A.1 shows validation R^2 across 30–50 Bayesian optimization trials. The optimizer converges within the first 15–20 trials, with subsequent trials confirming the region of best performance. Figure A.2 shows the parameter-level distributions: the learning rate concentrates around a narrow range, while L_1 strength shows a clear optimum that balances sparsity and model capacity.

A.2 Phase 2: Capacity

With training stability established, the second phase searches parameters controlling model capacity: embedding dimension, hidden dimension, lookback window, and dropout. Figure A.3 shows a modest improvement over the Phase 1 baseline, indicating that the model is not strongly sensitive to capacity parameters within the explored range. Figure A.4 reveals that

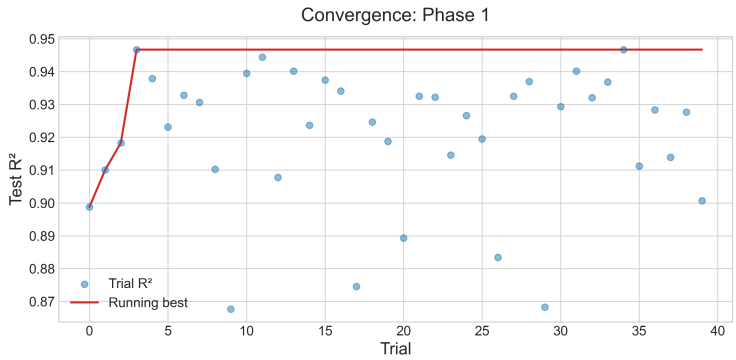


Figure A.1: Phase 1 (Stabilization): validation R^2 across Bayesian optimization trials.

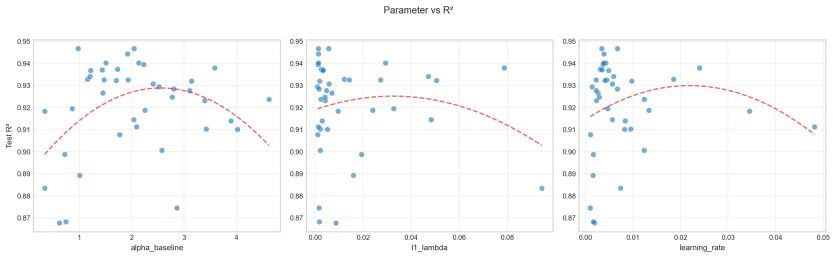


Figure A.2: Phase 1 parameter distributions: learning rate, L_1 strength, and baseline weight. Each point represents one Bayesian optimization trial, colored by validation R^2.

intermediate values of embedding and hidden dimensions perform best, consistent with the low-sample regime where excessive capacity leads to overfitting.

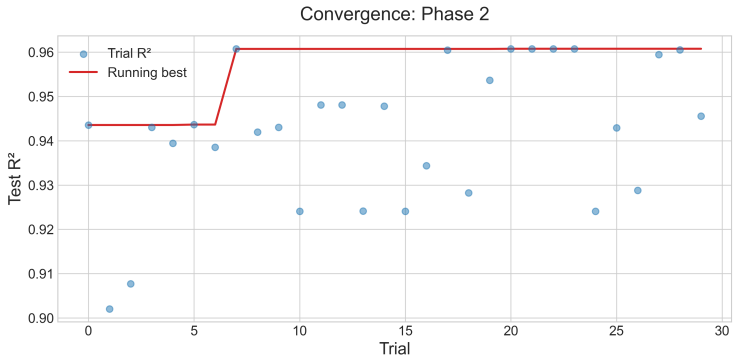


Figure A.3: Phase 2 (Capacity): validation R^2 across Bayesian optimization trials.

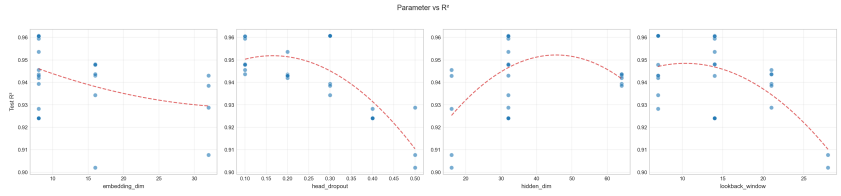


Figure A.4: Phase 2 parameter distributions: embedding dimension, hidden dimension, lookback window, and dropout. Each point represents one Bayesian optimization trial, colored by validation R^2.

A.3 Phase 3: Baseline

The final phase fine-tunes the baseline component: baseline weight, lag configuration, and initial bias. These parameters control how organic demand is modeled relative to media effects. Figure A.5 shows that the marginal improvement is smaller than in earlier phases, suggesting that the model is approaching a stable configuration. Figure A.6 shows the explored distributions; the baseline weight and lag configuration exhibit clear preferences, while the initial bias has a flatter landscape.

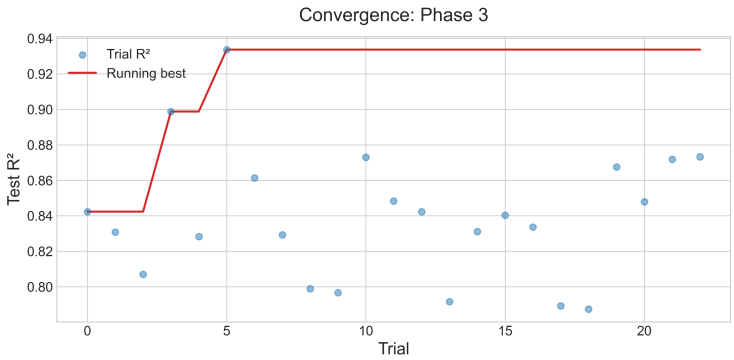


Figure A.5: Phase 3 (Baseline): validation R^2 across Bayesian optimization trials.

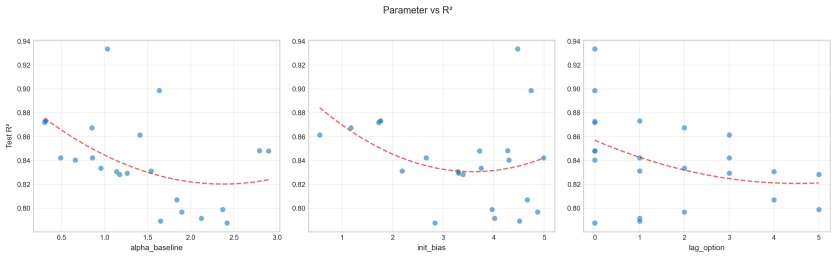


Figure A.6: Phase 3 parameter distributions: baseline weight, lag option, and initial bias. Each point represents one Bayesian optimization trial, colored by validation R^2.

Appendix B

Per-Merchant Temporal Analysis

This appendix reports the temporal analysis figures for all three evaluation merchants. The main chapter presents results for a single representative merchant; this appendix provides the complete set for reproducibility and cross-merchant comparison.

For each merchant, three figures are shown: the neural model's cumulative adstock curves, the Bayesian MMM's cumulative adstock curves, and the direct NNN-vs-MMM comparison for the top-3 channels by total spend.

B.1 Merchant 1

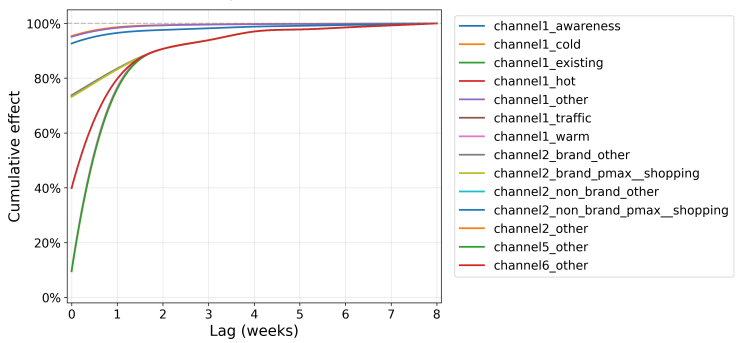


Figure B.1: Merchant 1 — Neural model: cumulative temporal response profiles by marketing channel. Each curve shows the cumulative attention weight as a function of lag in weeks, representing how the model distributes the effect of past spend over time. Steeper curves indicate faster-decaying effects.

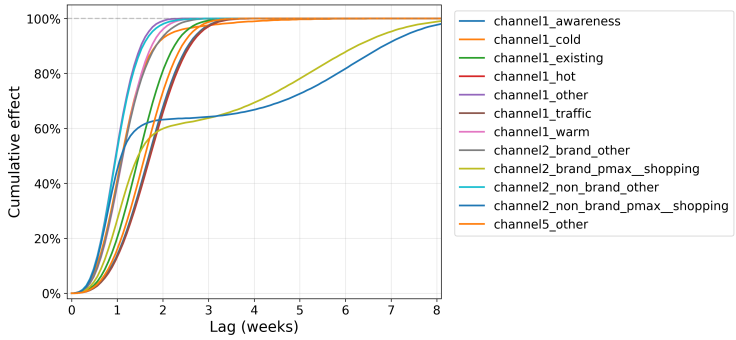


Figure B.2: Merchant 1 — Bayesian MMM: cumulative adstock curves by marketing channel. Each curve shows the cumulative carryover weight as a function of lag in weeks, derived from the fitted Weibull adstock parameters.

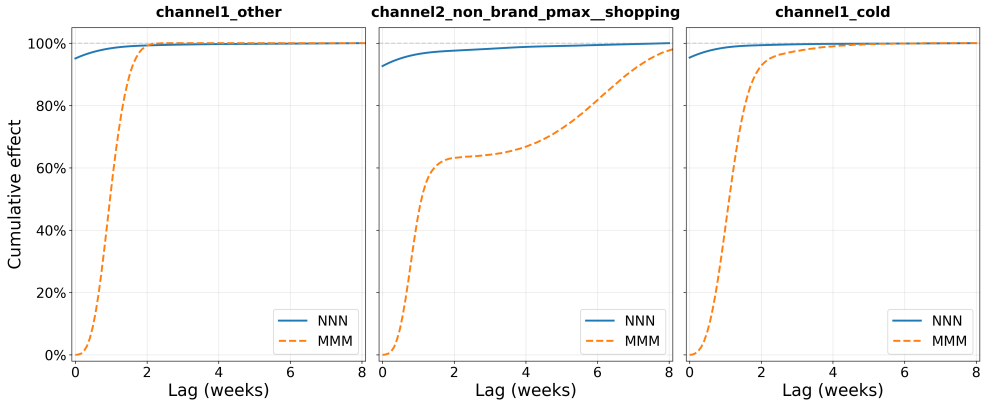


Figure B.3: Merchant 1 — Side-by-side comparison of cumulative temporal response profiles for the three highest-spend channels: neural model (solid blue) vs. Bayesian MMM (dashed orange). The neural model consistently saturates faster, concentrating the effect in the first 1–2 weeks, while the Bayesian MMM distributes carryover more gradually over longer horizons due to its parametric Weibull constraint.

B.2 Merchant 2

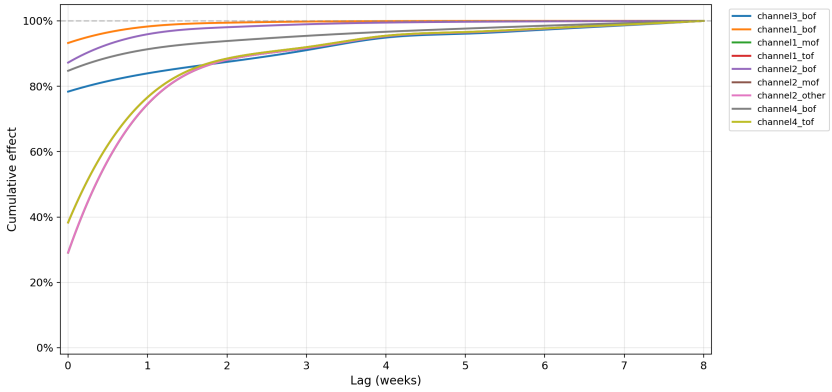


Figure B.4: Merchant 2 — Neural model: cumulative temporal response profiles by marketing channel. Each curve shows the cumulative attention weight as a function of lag in weeks, representing how the model distributes the effect of past spend over time. Steeper curves indicate faster-decaying effects.

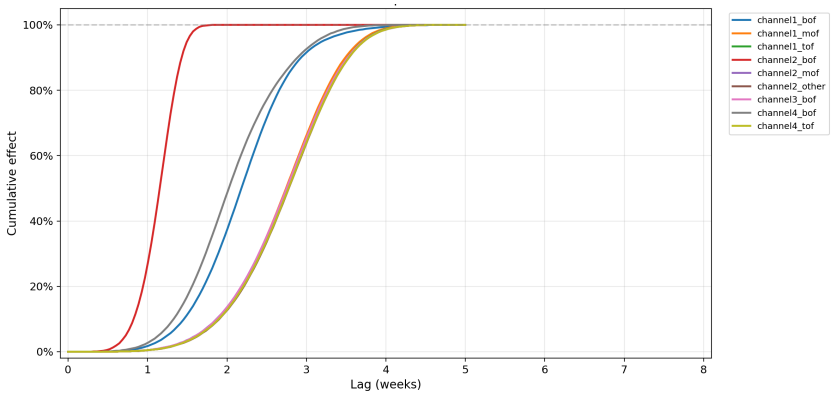


Figure B.5: Merchant 2 — Bayesian MMM: cumulative adstock curves by marketing channel. Each curve shows the cumulative carryover weight as a function of lag in weeks, derived from the fitted Weibull adstock parameters.

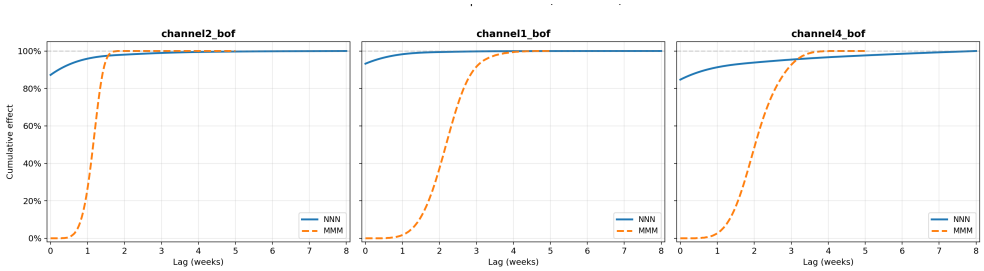


Figure B.6: Merchant 2 — Side-by-side comparison of cumulative temporal response profiles for the three highest-spend channels: neural model (solid blue) vs. Bayesian MMM (dashed orange). The neural model consistently saturates faster, concentrating the effect in the first 1–2 weeks, while the Bayesian MMM distributes carryover more gradually over longer horizons due to its parametric Weibull constraint.

B.3 Merchant 3

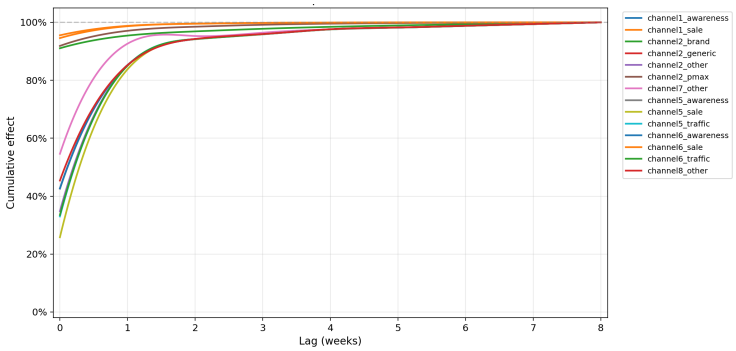


Figure B.7: Merchant 3 — Neural model: cumulative temporal response profiles by marketing channel. Each curve shows the cumulative attention weight as a function of lag in weeks, representing how the model distributes the effect of past spend over time. Steeper curves indicate faster-decaying effects.

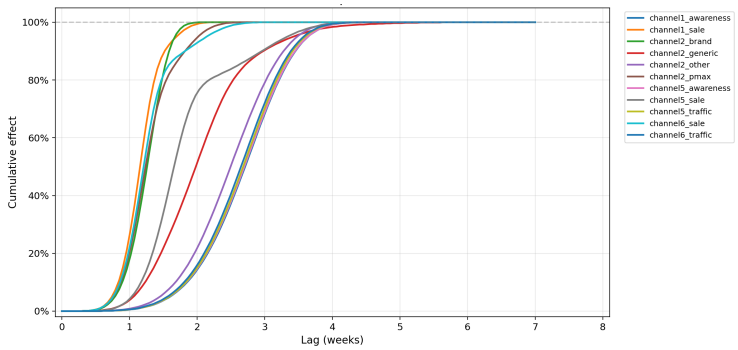


Figure B.8: Merchant 3 — Bayesian MMM: cumulative adstock curves by marketing channel. Each curve shows the cumulative carryover weight as a function of lag in weeks, derived from the fitted Weibull adstock parameters.

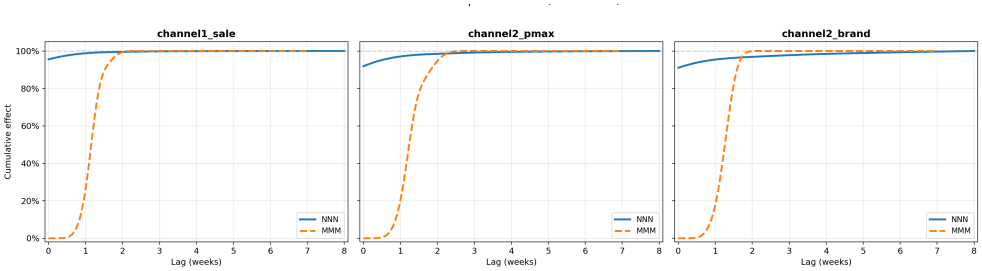


Figure B.9: Merchant 3 — Side-by-side comparison of cumulative temporal response profiles for the three highest-spend channels: neural model (solid blue) vs. Bayesian MMM (dashed orange). The neural model consistently saturates faster, concentrating the effect in the first 1–2 weeks, while the Bayesian MMM distributes carryover more gradually over longer horizons due to its parametric Weibull constraint.

TRITA-EECS-EX-2026:385
Stockholm, Sweden 2026

www.kth.se